

AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

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Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2-CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land-sea mask, WAM sea points) regenerated per leg.

Where we are. Eight real bugs and one architectural redesign (the coupled slab) later, the cycle is proven end-to-end with real cavities — awiesm3 year → PISM decade → awiesm3 year on the changed mesh, plus one further clean cycle. The 10-year production attempt then ran eleven more increments *with the cavities accidentally amputated by an overreaching mesh-hygiene fix* (discovered at wrap-up, reverted; those legs are tainted). The campaign continues, and the diagnosis has turned to physics: the Ross-front blowup is increment magnitude, not coastline fragility, and the increments are large because the PI-control cavity melt runs $\sim 7\times$ observations. That melt is traced to cavity currents $\sim 6\times$ too fast — and a standalone dissection (§1.13) then shows those fast currents are *injected by the mesh increment*: the geometry and the water are both fine (a cold start on the changed mesh settles to the healthy ~ 2.6 cm/s), but the *remapped restart is internally inconsistent with the new mesh* through `hnode` (the ALE layer thickness): the remap’s ALE-rescale step compresses the changed-column `hnode` using a stale `eta`, seeding a barotropic spin-up. Fixing that (nominal `hnode` at the changed columns) is a small `remap_hnode` edit and *halves* the over-speed (17→10 cm/s in isolation), but does not fully restore health: reaching ~ 4 cm/s needs nominal `hnode` over the *whole* ocean, which points not at the cavity fill but at the **vertical-coordinate seam** — `zstar` (stretched layers) outside vs. the rigid-lid cavity (nominal layers) inside. `linfs` (fixed layers everywhere) removes it: the raw remap under `linfs` does not spin up (decays to ~ 1.7 cm/s, no `hnode` surgery) — the clean complete fix, if a linearised free surface is acceptable model-wide (§1.13, §1.14). This is confirmed in the coupled chain, and then closed: with `linfs` plus a fill fix for a separate artifact (newly-iced columns inheriting warm open-ocean water, cured by seeding open→cavity columns from the nearest existing cavity), the `movcav37` run carries the full decade 1900–1909 — **ten coupled years, nine mesh increments, cavity current held at 1–2 cm/s throughout, no spin-up and no blowup**, cavity preserved, melt in the observed range (§1.15, §1.16). Rolling the coupled slab out to the fixed-mesh siblings then caught a months-old silent bug — `develop/develop-cc` never set the OIFS NAMMCC switches, so the coupled ice thickness was discarded and their Arctic went ice-free every summer; fixed in config, the 10-year sea-ice climate validates against an independent CMIP7 spin-up (§1.18). The mesh-increment crash family (`voskin:282`, Antarctic) is now inventoried and instrumented, reproduction pending (§1.19). Of the two further crash faces that instrumentation exposed, the leg-start one is **solved**: not the SSH solve it looked like, but **Bug VIII** — `fer_solve_Gamma` publishing an unassigned automatic-array element into the GM streamfunction wherever a node’s surrounding elements invert its solve range, giving a 10^{251} bolus velocity that detonates the tracer solve. A cavity front on steep bathymetry is what creates that node geometry, which is why a decade of FESOM never hit it. With it fixed the cycle then ran itself from year 1906 to **1924** — **18 further coupled years, zero blow-ups** — and the system is visibly equilibrating: shelf area flattening, cavity melt settling at 0.8–0.9 m/yr inside the observed band, and the per-cycle mesh increment decaying $430 \rightarrow 45$ nodes removed (§1.21).

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1 The campaign, in the order it happened

Each subsection carries the date it was worked. The bug numbering (I–V) is the order of *discovery* and is kept as the canonical naming.

1.1 The restart seam and the cold-start control (2026-07-08–13)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	1200 s	cold start	stable, CFLz 3.44
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n , 0 CFLz)

The ocean lives happily in PISM’s cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are **TAINTED** (measured under Bugs I+II) and the finding that matters survived into the fixes below: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

1.2 Bug I — PISM was fed surface water as ice-base forcing (2026-07-14)

`coupling_ocean2pism` built PISM’s `-ocean` forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-sellevel,150/600` sat in dead code). PISM received ice-base water up to +6.5°C even from a healthy ocean, melted at ~ 30 m/yr, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM’s own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

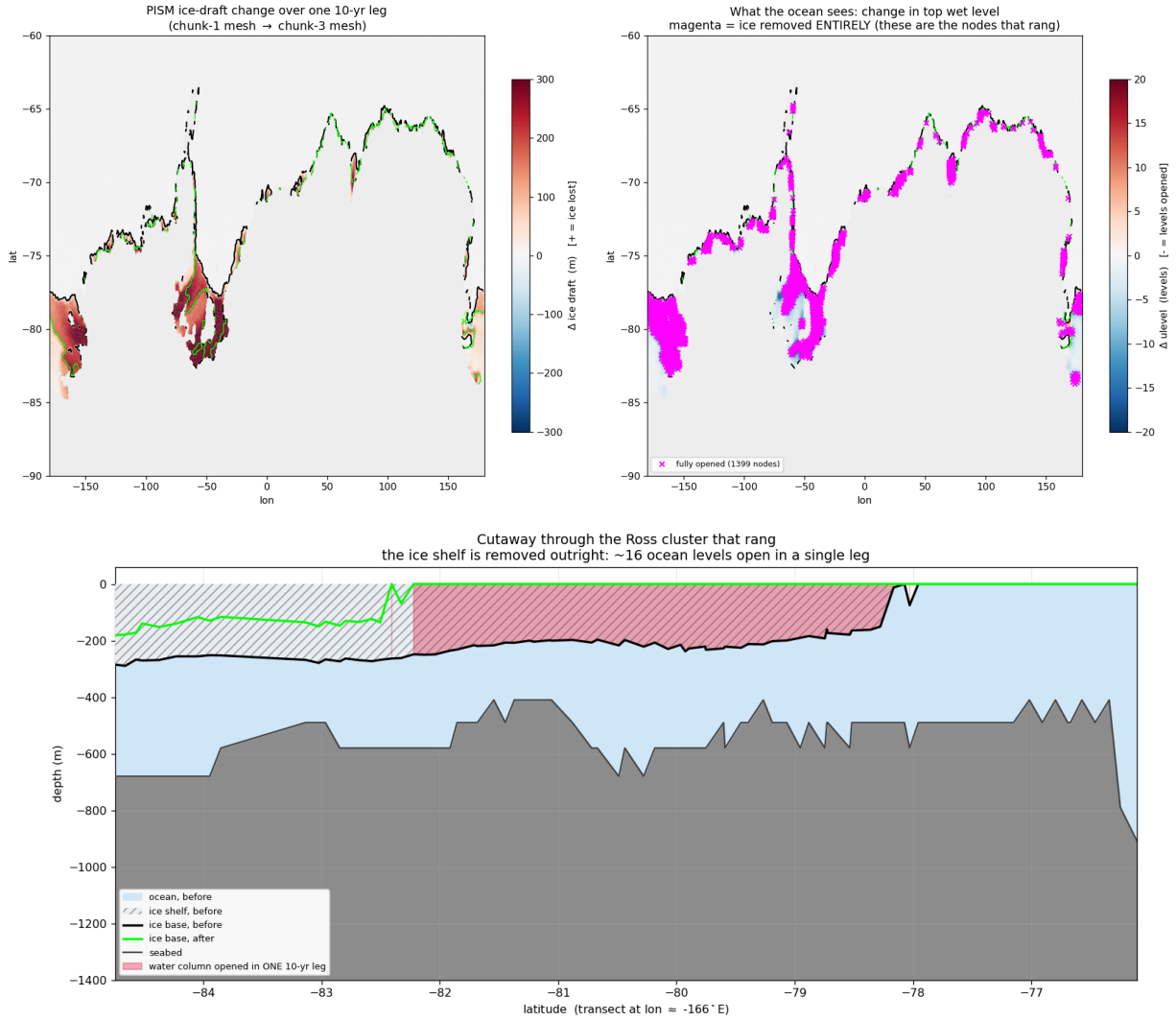


Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats ≈ 440 km (section: ice base $-200 \dots -280$ m $\rightarrow$ open ocean across four degrees of latitude).

Fix (93a207ee): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean given: fw→shelfbmassflux ( $\times 1000$ , sign preserved), sst→shelfbtemp`. One model owns the ice-ocean interface. The `-ocean given` machinery existed all along but silently fell back to `th` because its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: 3f07d7a2, 74671c27, bb9d00c2).

1.3 Bug II — the exchange weights scrambled the coupled ocean (2026-07-14)

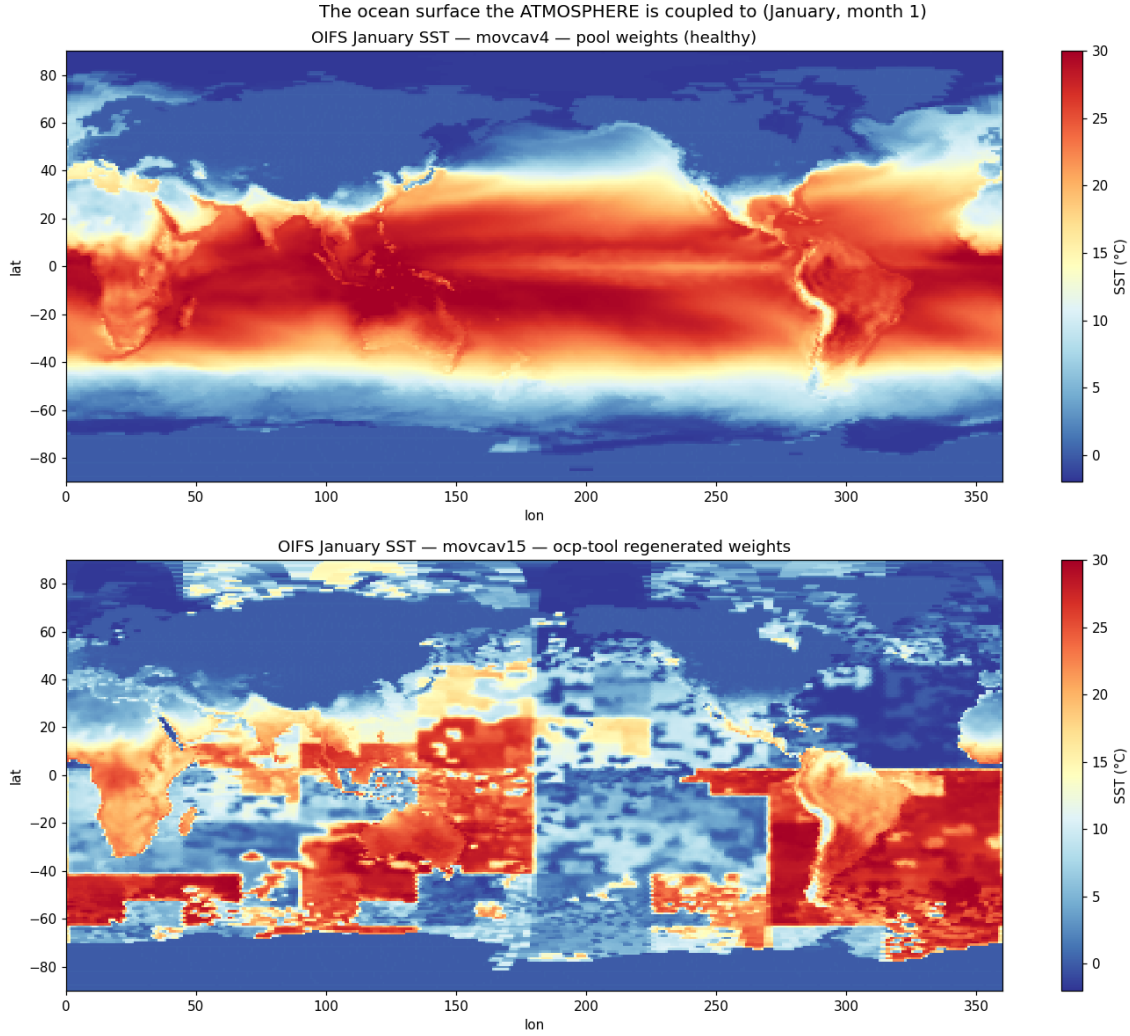


Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

FESOM exchanges coupling fields in *partition* (*my_list/rank*) order; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out* order (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (c21fe7c2): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_<nproc>/rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at 42–58°, correct ones pass at $\leq 0.025^\circ$).

1.4 Bug III — the OASIS coupling restart was compound-scrambled (2026-07-14)

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *runtime*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the

restart’s own surface temperature: +0.15). OIFS’s first coupling interval saw 305 K water under polar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (abe56ee2): un-permute via the old mesh’s `rpart.out` → gather → re-permute via the new mesh’s; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.5 The wave model’s frozen coastline (2026-07-15)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM’s Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM’s sea-point set was rebuilt from ETOPO1 with the southern limit extended $78 \rightarrow 85^\circ\text{S}$, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened ($29\,555 \rightarrow 30\,832$ sea points), 283 of them south of 55°S : exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

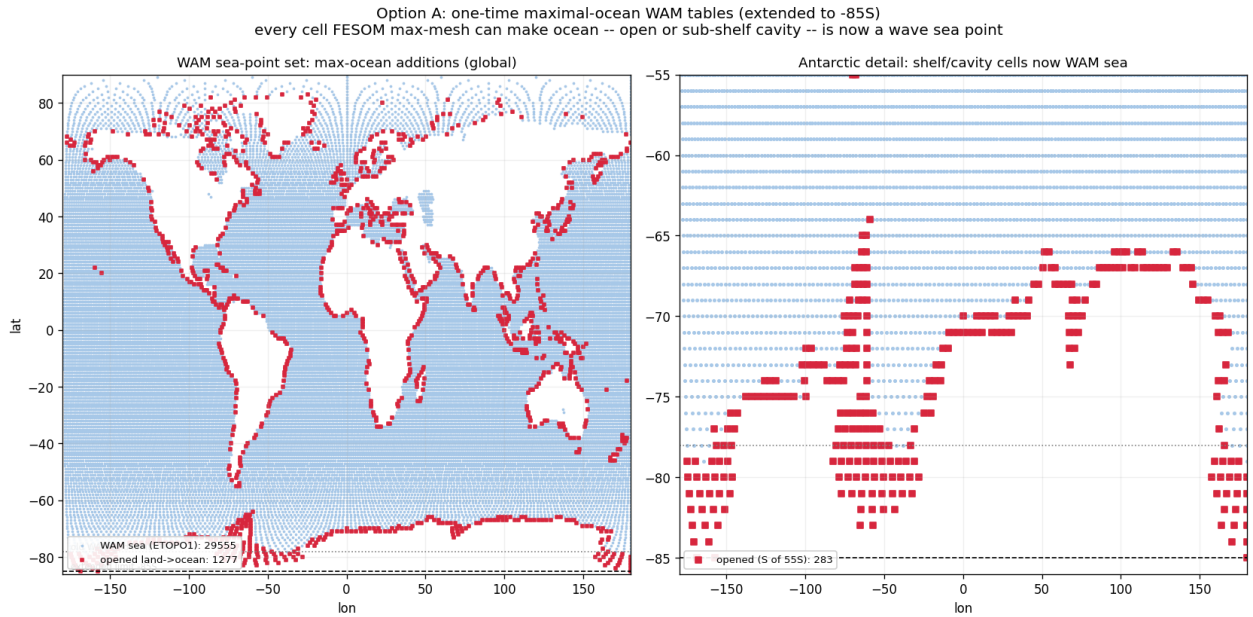


Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land→ocean because the FESOM max-mesh has a water column there. Pre-registering them closes the WAM-Stokes route into `voskin:282`. (It does *not* close the line entirely: `voskin:282` later recurred through a different, in-step route — §1.19.)

1.6 Bug IV — the ice-surface-temperature coupling is architecturally unstable (2026-07-14→16)

This is the mid-run OIFS crash family (`cubasen/vsurf/voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS’s ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted `ist` (`ist·a_ice ÷ ci` on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived $3\times$ longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM’s skin solve relaxes toward $t^* = T_f + a2ihf \cdot zsniced/con$ with the received flux *frozen* for the hour and no dQ/dT term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to –954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = a2ihf + \lambda(t_{ref} - t)$ with $\lambda = 4\varepsilon\sigma t_{ref}^3 + \lambda_{turb} \approx 20 \text{ W/m}^2/\text{K}$ (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, 38a558b5). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere’s ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1 \text{ kW/m}^2$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean -15.8 kW/m^2 below 150 K (100% negative), -4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of ~204 K under ~275 K maritime air**; the sensible + latent terms then reach $+17.8 \text{ kW/m}^2$ per tile area. With weighted-`ist` the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOUPL=.true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere’s surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM’s oscillating solve → remap → *pinned* OIFS skin → kW fluxes against an unresolvable surface → remap → FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped between completing a year and dying at day 2 (runs `movcav25–31`: raw/weighted $\times \lambda = 3.4/20 \times$ dirty/clean initial state — every combination crash-prone).

Root fix: the coupled slab (§1.8). Stop prescribing the atmosphere’s ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.7 Bug V — iterative coupling never handed over to PISM (2026-07-16)

The awiesm3→PISM model switch is resolved at SimulationSetup init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of `general.base_dir` before `esm_parser` resolves variables*. A `base_dir` containing `${user}` was probed verbatim; the silent `isfile()` miss fell back to “chunk 1, model 1” — awiesm3 self-resubmitted to its own final date and PISM never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal `base_dir` (movcav16 era) and to `['fesom', 'oifs', ...]` under `${user}`.

Fix (d61297e8): `_early_resolve()` expands `${...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine awiesm3→PISM handover of the campaign (movcav27), and a full awiesm3→PISM→couple_out→chunk-3-couple_in chain (movcav29).

1.8 The coupled-slab architecture (2026-07-16/17)

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state**. OIFS’s surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent dQ/dT , finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM’s albedo still drives the radiation; FESIM’s own `ist` solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut by construction. The machinery is EC-Earth/KNMI heritage (`ICESTATENEMO`, 2008; the `SI3` branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε -guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
LNEMOSICOU	.false.	vlamsk: sea-ice tile skin conductivity becomes <i>finite</i> (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
LNEMOLIMTEMP	.false.	disables the hourly overwrite of OIFS’s 4-layer ice temperature (SP_SB YTL) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM’s ist send becomes inert.
LNEMOLIMTHK	.true.	activates the callpar fill of ZTHKICE/ZSNTICE from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
LMCCDYNSEAICE	.true.	master enable; false force-resets the whole LNEMOLIM* family (sumcc).
LNEMOLIMALB	.true. (default)	FESOM’s albedo drives OIFS radiation (surfrad_layer → FESIM getter; raw–raw convention, consistent).
ECE_CPL_NEMO_WEIGHTED_ICE	.true. (inert)	only acted inside the disabled ist ingest.
FESOM latm_owns_ist	.true.	FESIM growth budget takes the atmosphere flux directly (Qatmice=a2ihf, the ECHAM convention); qres/qcon/qlam booking off.

The sumcc guard “LNEMOSICOU without LNEMOLIMTEMP does not make sense” correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS’s ice surface was a mirror bolted to whatever FESOM’s oscillating solve sent; now it is a real solved surface conducting through FESOM’s actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a ZMASK threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \dots -80 \text{ W/m}^2$ (textbook polar values, was -10 kW), and FESIM’s internal temperature tracks its anchor to $< 0.1 \text{ K}$.

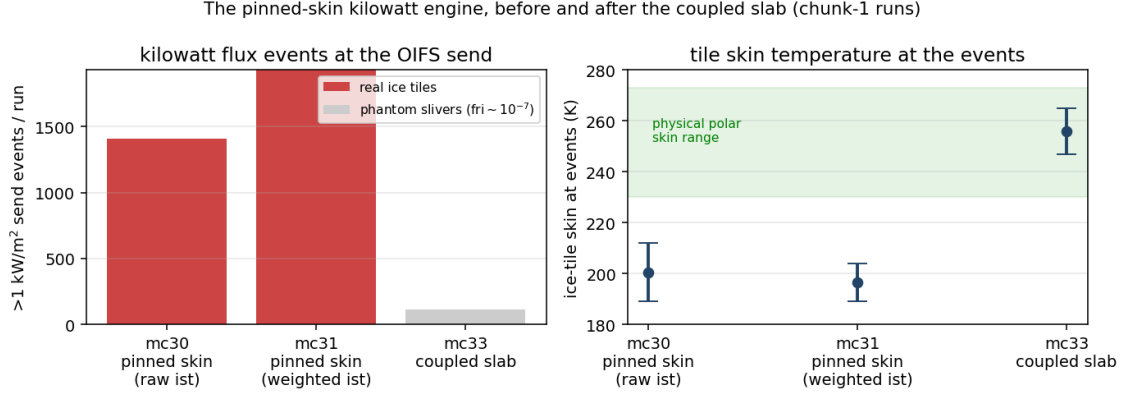


Figure 4: The kilowatt engine, before and after. Left: $>1 \text{ kW/m}^2$ events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

1.9 First fruits of the clean decade: cavity melt against observations (2026-07-17)

The chunk-1 year that forced the first clean PISM decade, split by sector and set against the satellite record (Rignot et al. 2013; Adusumilli et al. 2020):

sector	observed (m ice/yr)	movcav33 chunk-1 (m w.e./yr)		
		mean	median	nodes
Filchner–Ronne	0.3–0.4	0.89	0.20	1085
Ross	~ 0.1	0.90	0.31	1075
Amundsen–Bellingshausen	14–27 (PIG 16)	7.8	6.4	229
Antarctic Peninsula	0.4–3.8	7.5	4.4	186
Wilkes (Totten sector)	4.7–9.1	8.3	6.4	296
Amery	0.6–0.7	4.0	3.3	128
E. Weddell / Dronning Maud	0.3–0.9	4.4	3.5	287
circum-Antarctic	0.8 ± 0.7	2.84	0.79	3286

The cold/warm dichotomy that dominates the observations is reproduced: the cold-cavity giants sit at 0.2–0.3 m/yr medians while the warm sectors run 4–8, and 13% of the cavity area refreezes (marine ice, as observed under Ronne/Amery). Residual biases: the East Antarctic coastal sectors (Dronning Maud, Amery) run $\sim 5\times$ warm, and the Amundsen runs $\sim 2\times$ cold — both worth revisiting once multi-year coupled statistics exist. Caveats: node statistics (unweighted), m w.e. vs the observations’ mice ($\times 0.92$), one coupled year, and PISM’s initial geometry carries reduced versions of the big shelves. For scale: the jigsaw-era coupling produced 28 m/yr.

1.10 The increment test crashes, twice — and passes (2026-07-17/18)

The first chunk-3 leg — awiesm3 on the PISM-changed 208 438-node mesh, stage 6/6 — absorbs the mesh change and runs for weeks of model time before FESOM dies of an SSH explosion (η jumping $\pm 40 \text{ m}$ in one step at the Antarctic Peninsula tip; day 19–51 depending on MPI reduction order — the state is metastable). The autopsy was cheap for two reasons: the leg costs 3–4 minutes of wall time, and a temporary XIOS file group with `sync_freq="1d"` flushes daily fields to disk as they complete, so even a crashed run leaves full evidence (default XIOS files sync only on close: 51 days of output, zero records). The hunt found *two independent bugs*, stacked at the same place.

Phantom zero-water at draft-changed cavity columns --- George VI / Marguerite Bay
black outlines: elements whose cavity draft the mesh change touched

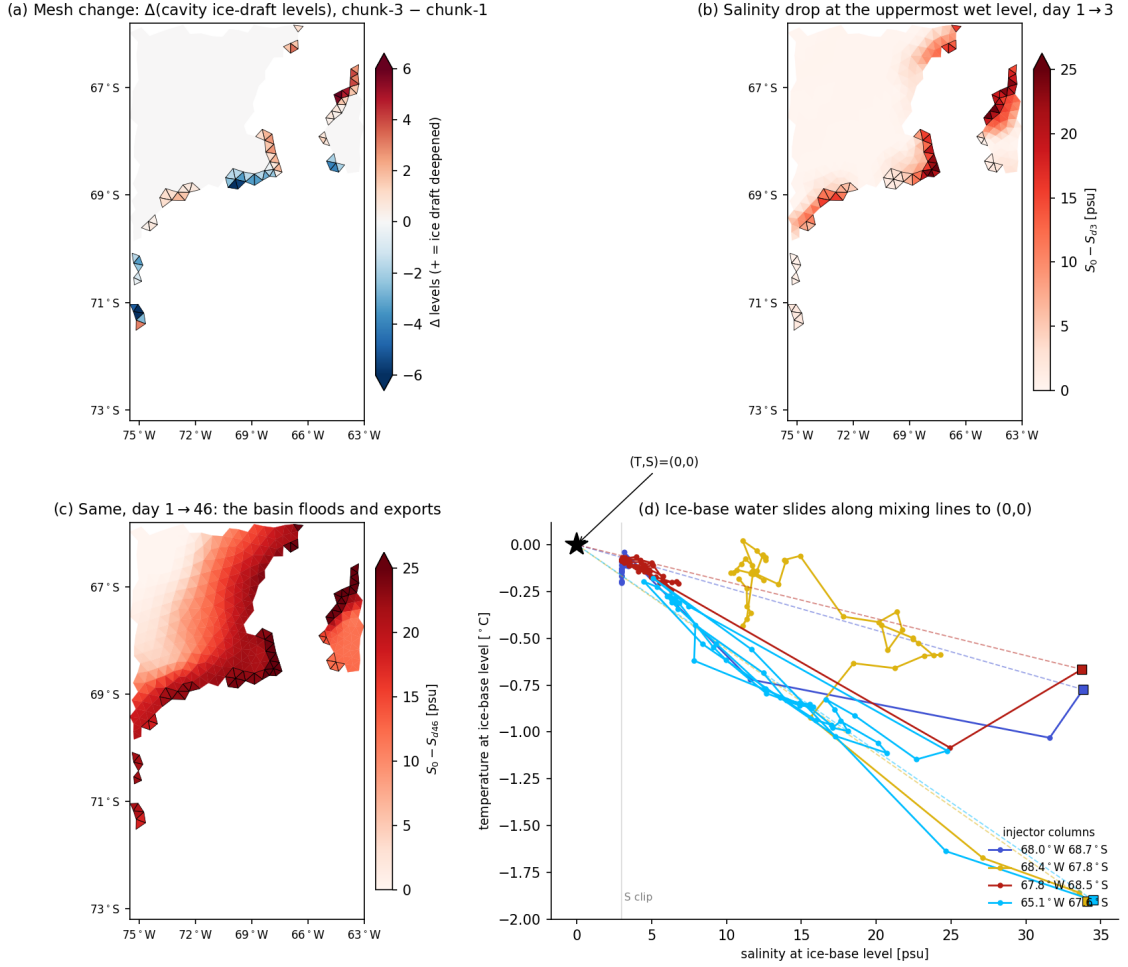


Figure 5: Bug VI seen in one basin (George VI / Marguerite Bay; black outlines = the 73 elements whose cavity draft the mesh change touched). (a) The mesh change: ± 1 –6 cavity levels. (b) By day 3 the uppermost-wet-level salinity has dropped up to 25 psu *exactly on the outlined elements*. (c) By day 46 the basin has flooded and exports along the coast. (d) The fingerprint: (S, T) at the ice-base level slides along the mixing lines toward $(0, 0)$ with *identical* T - and S -replacement fractions — not a flux (surface freshwater is $50\times$ too small and negative), but a *multiplicative drain*: both tracers lose the same fraction of their value every step.

Bug VI — the layer-thickness mismatch drain

The remapped restart carries `hnode` inherited from the old mesh's `zstar` state (e.g. 9.964 m where a formerly ice-free column scaled its layers with η). At runtime, cavity columns and their open-ocean edge ring (`ulevels_nod2D_max > 1`) keep layer thicknesses *fixed at the mesh nominal* (10.000 m) and are excluded from the per-step `hnode ↔ hnode_new` sync — so the mismatch is immortal. FESOM's tracer T^* update applies values $\cdot (\text{hnode} - \text{hnode_new}) / \text{hnode_new}$ every step: a -0.36% /step multiplicative drain on T and S . Per-operator probes at a Getz injector column measured -0.1226 psu/step against a predicted $33.64 \times 0.0036 = -0.1224$; an unchanged FRIS control column measured exactly zero. At free-surface nodes next to the cavity edge the drain even *accelerates*: η grows, the nominal-vs-restart gap grows with it. **Fix** (FESOM, `restart_thickness_ale`): after the restart read, force `hnode` at the whole excluded class back to the fixed nominal and rebuild the depth arrays. Validated by

intervention: the drain vanished (+0.0002/step), day-1 fields healthy.

Bug VII — the drowned island

With Bug VI fixed the run survived longer but still exploded at the Peninsula tip, η climbing a steady +0.11m/day. The reason is geometric, not numeric (Figs. 7, 6): the ice2fesom regrid remapped PISM’s *categorical* mask **bilinearly** — a few-cell island surrounded by mask-4 ocean smears to $\approx 3.x$, rounds to ocean, and is deleted from the submesh. PISM’s own bed there is +38 to +218m above sea level: bedrock island, regardless of ice. (The island entered the campaign *bare* — the spinup state carries 0.0m of ice on all ten cells; a thin 0.1–0.4m cap has been *growing* during the coupled decades. A second, independent deletion path was found while examining it: PISM encodes ice-free bedrock as mask=0, which the node classifier silently defaulted to ocean; fixed — bare bedrock above sea level is land.) The ocean then drives a runaway coastal jet where a coastline should be. **Fix** (ice2fesom): remap the categorical mask nearest-neighbour; rebuild the mask=5 outside-domain marker (which `remapnn` extrapolation would erase) from a bilinear domain indicator. Continuous fields stay bilinear.

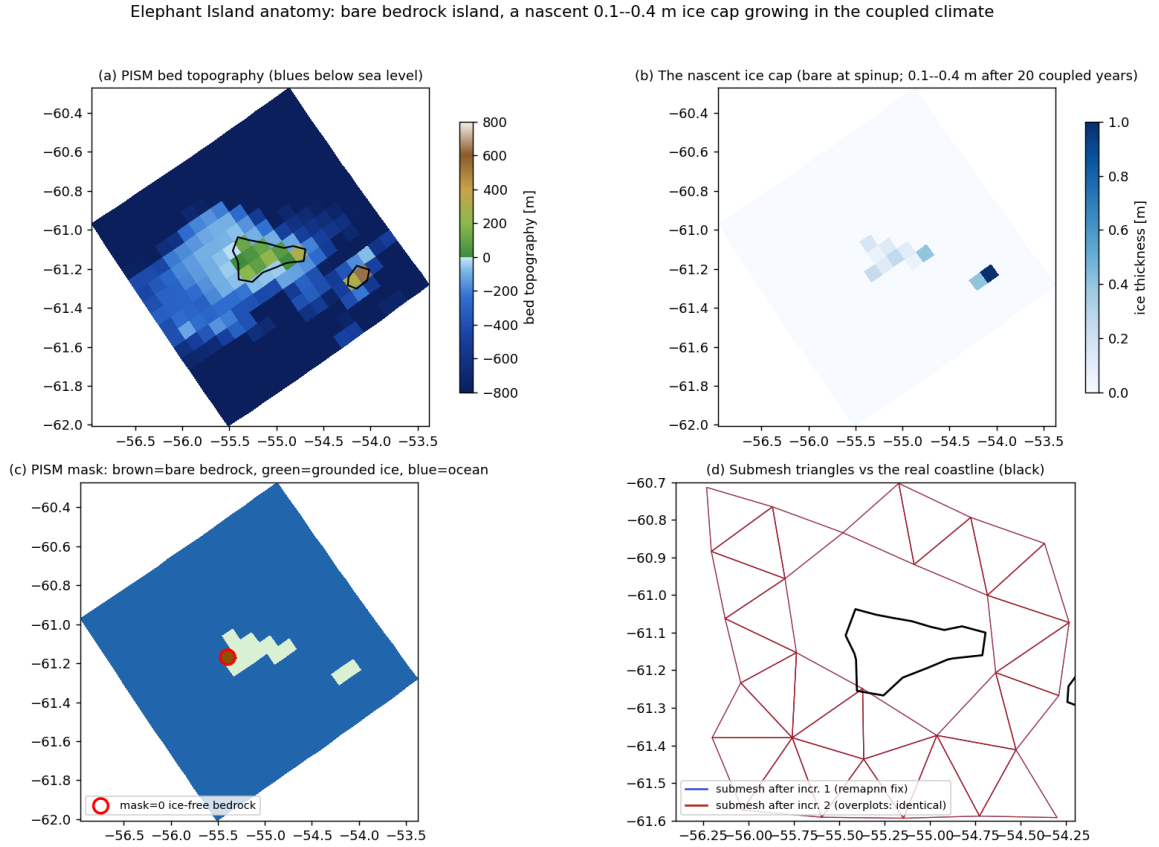


Figure 6: Elephant Island anatomy. (a) PISM bed topography, ETOPO-style colors: a genuine bedrock island (+4 to +297 m) on its shallow platform, Clarence’s peak to the southeast. (b) The *nascent* ice cap: bare at spinup, 0.1–0.4 m grown over 20 coupled years (DEBM accumulates here). (c) The PISM mask; the red circle is the single mask=0 (ice-free bedrock) cell that exposed the classifier hole. (d) The submesh renders the island identically after increments 1 and 2 — the nearest-neighbour fix holds.

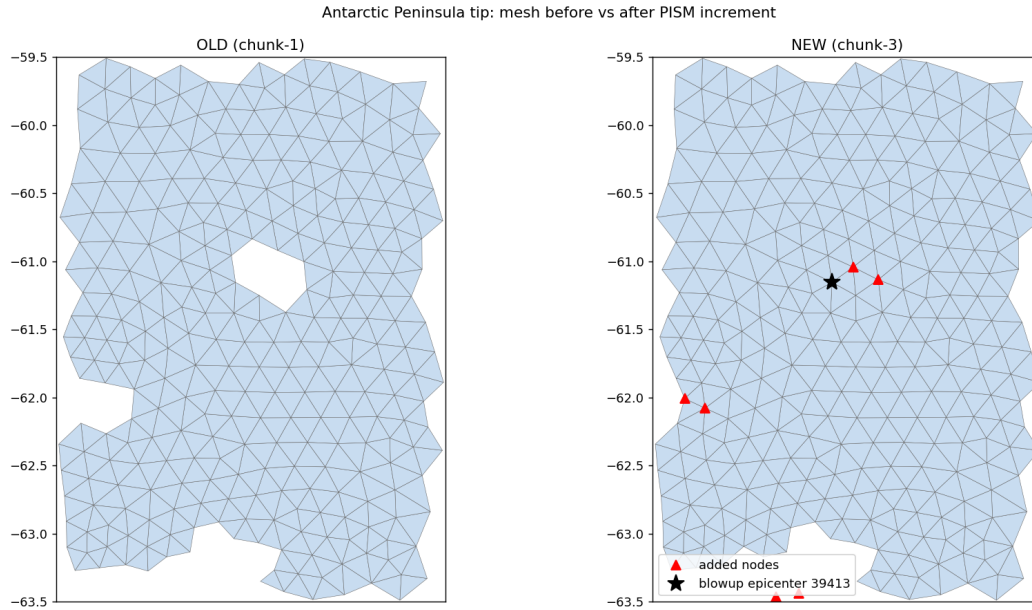


Figure 7: Bug VII. Left: the chunk-1 mesh has Elephant Island (the hole at 55.4°W , 61.1°S) and the Clarence/Gibbs notch. Right: after the PISM decade the bilinear mask remap has rounded both islands to ocean — the added nodes (red) fill them in, and the blowup epicenter (star) sits exactly on the drowned island.

Ruled out by measurement (each was a plausible suspect): surface freshwater fluxes; OIFS runoff/calving and the ECE_LANDICE snow cap; the OASIS `feom` masks; the remapped T/S state (freezing cap ran: 1137 columns); mesh-file level-range consistency (zero violations); the zero-filled dry restart levels (padding them — kept as a permanent hygiene pass — reproduced the crash bitwise); and the vertical-advection top boundary (cavity guards in both the implicit and explicit schemes changed nothing — see §4). The couple_in warning about a missing `cap_new_cavity_temp.py` is a red herring: the freezing cap runs as Fortran inside the remap binary.

The increment test passes

With both fixes, `movcav33` chunk-3 completed the full year 1901 on the changed mesh (25 min wall, no blowup; the Peninsula node’s SSH wanders seasonally instead of pumping) and handed over to the chunk-4 PISM decade unattended — **stage 6/6 closed**. The long run `movcav34` (10 awiesm3 years / 100 PISM years) then repeated chunks 1–4 cleanly on the *minimal* build (hnode fix only; the advection guards were reverted after being falsified). Fixes are committed: FESOM `awiesm3-implicit-ice-surftemp`, `esm_tools development_interactiv_mesh_awiesm3`.

The next problem: the second increment and the hygiene overreach (2026-07-18)

`movcav34`’s chunk-5 (year 1902, after 20 PISM years) blew up at day 16 at the western Ross front, where the second increment opened cavity columns wholesale (ulev 13 $\rightarrow$ 1) amid a 506-node ice advance. The new coastline there was structurally rotten: 24 nodes within 1.5° of the epicenter held by ≤ 3 elements, including an 18-levels-under-ice cavity column held by exactly two triangles. A *coastline-hygiene sweep* was added to the submesh tool (iteratively remove ice-domain nodes held by ≤ 2 elements) and the chain then ran unattended through year 1911 and 120 PISM years — eleven further increments without a FESOM blowup, ending only in a recurrence of the old OIFS `voskin` overflow at chunk 13.

The overreach. At wrap-up the evolution diagnostics exposed the truth (Figs. 8, 9): the

hygiene sweep had *cascaded through the narrow cavity bands* — fringing shelves and cavity margins are exactly the few-element-wide structures the criterion eats rim-inward — and every post-1902 submesh carries only 14–111 cavity nodes instead of ~ 2600 . The node classifier was healthy (2223 shelf nodes flagged); the lake floodfill removed only 35 elements; the sweep itself did the damage. **The eleven “stable” legs ran with the ice-shelf cavities amputated — stability by amputation, and every result from movcav34 after chunk 4 is **TAINTED**.** The sweep was reverted the same day (commit ca0641b2 reverts 680daa55).

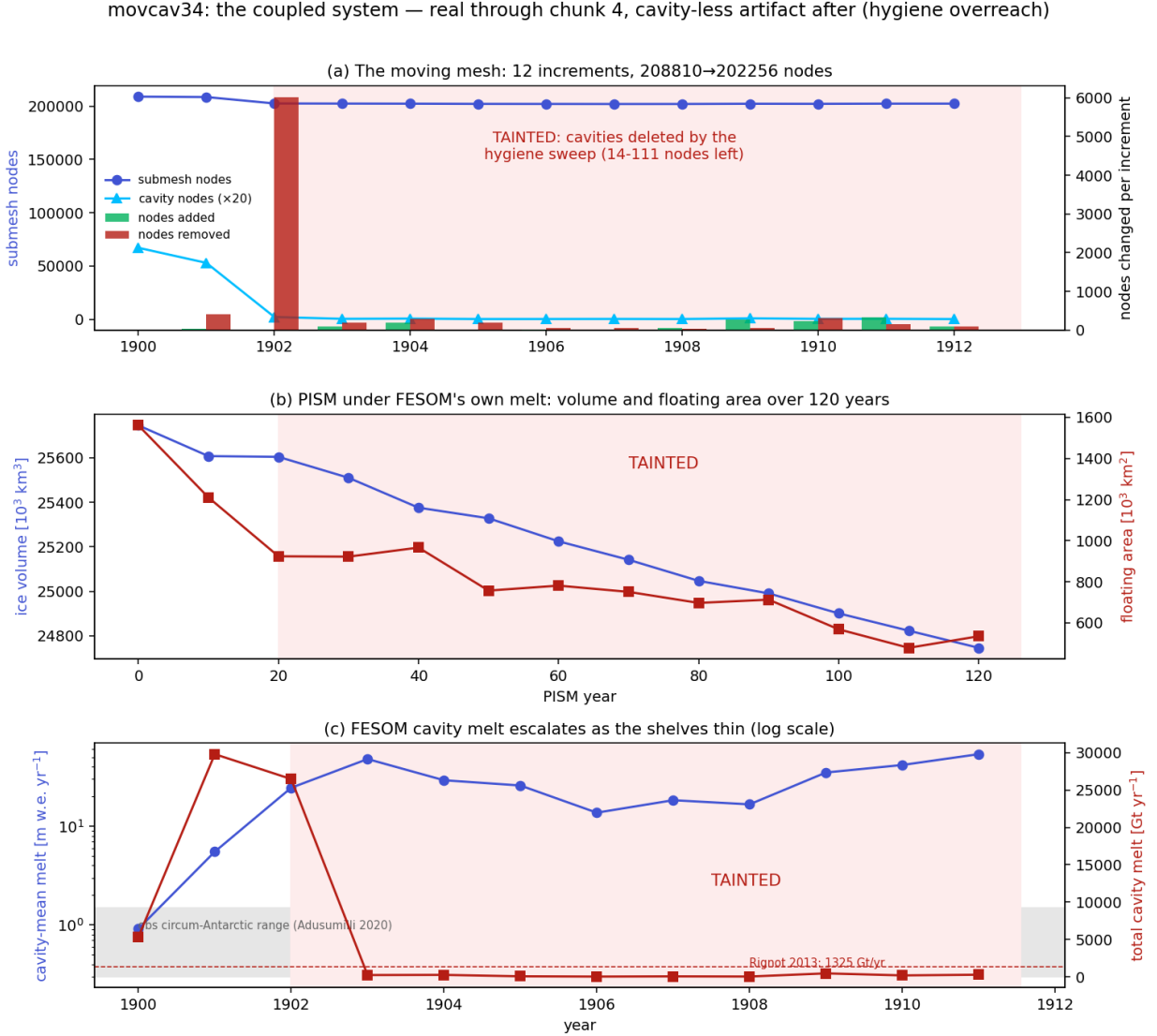


Figure 8: The movcav34 production run, honest version. (a) Submesh and cavity-node counts per increment: the collapse from 2647 to 111 cavity nodes at the 1902 mesh is the hygiene amputation. (b) PISM volume and floating area over 120 years: the decline is real coupled melt through year 20, then continues under forcing computed from an essentially cavity-less ocean. (c) Cavity-mean and total melt per leg (log scale): the escalation beyond ~ 1902 samples only the few surviving cavity slivers. Red shading = tainted.

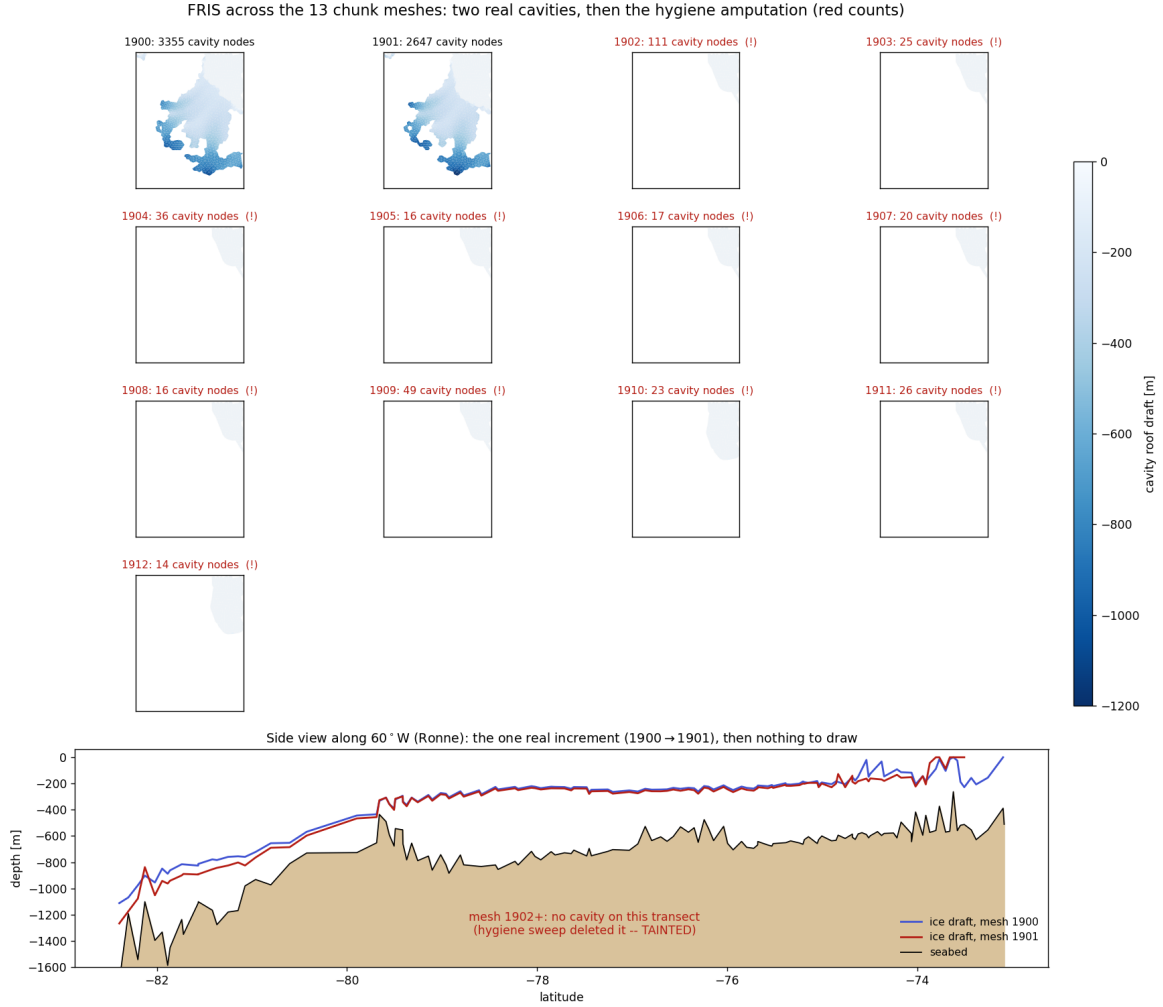


Figure 9: FRIS across the thirteen chunk meshes. Top grid: cavity elements coloured by roof draft — two real cavities (1900, 1901), then eleven empty frames with red node counts. Bottom: ice draft along 60°W for the one genuine increment (1900→1901; the southern draft deepens by 50–100 m), after which there is nothing left to draw.

The problem to solve next (Figs. 8, 9 are its statement): make the mesh increments survivable *without* destroying the cavities. What is genuinely proven and stays: the full cycle awiesm3 → PISM → awiesm3-on-a-changed-mesh works end-to-end with real cavities (movcav33 chunk-3 + handover; movcav34 chunks 1–4 reproduce it on the minimal build). The work queue, in order: (i) a cavity-preserving fragile-sliver treatment — removal must not disconnect or cascade through the few-element-wide cavity bands (guard on the shelf flag, a connectivity-preserving criterion, plus a safety valve that aborts the leg if the submesh cavity count drops far below the classifier’s flag count); (ii) rewind movcav34 to chunk-5 and re-run the increments with real cavities; (iii) the recurring OIFS voskin coastal overflow; (iv) the iterative-coupling loop ignores `final_date`; (v) the escalating warm-bias melt that drives PISM’s fast geometry increments in the first place. Items (i) and (v) are worked below.

1.11 The coastline is not the problem — the increment is (2026-07-18)

The Ross-front blowup looked like a fragile coastline (24 nodes held by ≤ 3 elements within 1.5° of the epicentre, including an 18-level cavity sliver on two triangles), so the first response was a hygiene sweep removing ice-domain nodes held by ≤ 2 elements. It ran the chain to year 1911 — but by *amputating the cavities* (Bug “VIII”, above). An offline harness on the saved 1902 flag files then settled the question by building submesh variants without submitting jobs:

- **Fragility is normal, not pathological.** The meshes that ran clean full years carry the *same* fragile census as the crashed one: pool 3470 nodes with ≤ 2 elements (429 in cavities), submesh_1901 3448 (378), versus 1902’s 3444. A control that kills the hypothesis: if ≤ 2 -element slivers were fatal, 1901 would have died too.
- **The increment is the violence.** Measured as per-node cavity-roof change against the previous submesh: increment 1 (pool→1901) max $|\Delta\text{draft}|$ 451 m, mean 0.96 m; increment 2 (1901→1902) max **1196 m**, mean **8.7 m** — $9\times$ the mean, opening 13 vertical levels at the epicentre in one decade beside a 506-node grounding advance. This is downstream of item (v): the warm-bias melt escalates (cavity-mean $0.9\rightarrow 5.5\rightarrow 24$ m/yr), so each decade’s geometry step is larger than the last.

Fixes committed. (a) A *cavity-amputation abort guard*: if the reduced submesh retains $< 50\%$ of the classifier’s flagged cavity nodes, the leg fails loudly instead of running a formally-valid but cavity-less ocean (the tripwire that would have caught Bug VIII in one couple_in instead of eleven tainted chunks). (b) The PISM mask-value 0 (MASK_ICE_FREE_BEDROCK) is now classified *land*: a second, independent island-deletion path (bare bedrock defaulting to ocean) that would have re-drowned Elephant Island once its nascent cap finishes melting. **Rejected:** a per-increment cavity-roof *rate limiter* (clamp the roof to within 75 m of the previous leg). It worked offline (retained 2221 cavity nodes vs 1864 unclamped) but was *rejected on physics grounds*: it makes the ocean’s cavity geometry deliberately lag the ice sheet’s, a coupler-imposed inconsistency between components. The increment must be tamed at its source (the melt, next) or by numerical treatment of the post-increment transient — not by distorting the coupled geometry.

1.12 Why the cavity melt is high and rising, and how to slow it (2026-07-18)

Item (v), the disease under the increments: the PI-control cavity melt runs at ~ 5.5 m/yr area-averaged against an observed circum-Antarctic ~ 0.8 m/yr, and *escalates* ($0.9\rightarrow 5.5$ m/yr over 1900→1901 on real cavities). Diagnosed from the monthly temp/salt/fw/unod/vnod output (Fig. 10); the cause is not where the first pass guessed.

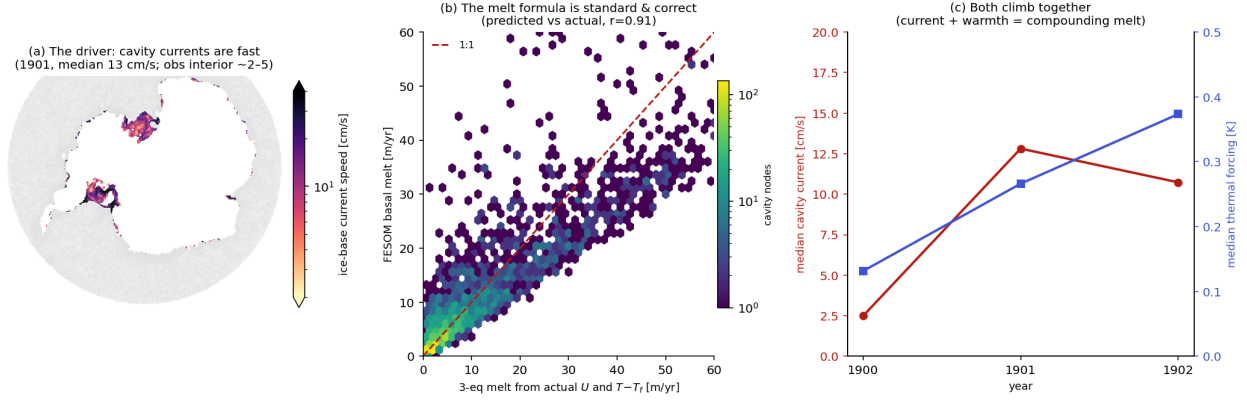


Figure 10: Why cavity melt is high and rising. (a) Cavity ice-base current speed (1901): median 13 cm/s, mean 18, 61% of nodes above 10 cm/s — fast, concentrated at the Amundsen/Bellingshausen shelves and the Ross/Filchner ice fronts (observed cavity interiors are a few cm/s). (b) The melt *formula* is standard and correct: fed FESOM’s actual currents and thermal forcing, the three-equation scheme reproduces the model’s own melt (predicted vs actual $r = 0.91$). (c) Both drivers climb together — thermal forcing (median $+0.13 \rightarrow +0.37$ K, warm-water intrusion) and current speed — so the melt compounds.

The chain, measured. Thermal forcing $T - T_f$ at the ice-base level is *near-realistic* (area-weighted $+0.14$ K; median $+0.13$ in 1900 rising to $+0.37$ in 1902). The Holland & Jenkins three-equation scheme in `cavity_param.F90` is standard (drag $a_k = 2.5 \times 10^{-3}$, $\gamma_T \approx 10^{-4}$ m/s), and fed the model’s own currents and thermal forcing it *reproduces the melt* ($r = 0.91$, 18 vs 16 m/yr): the formula is not miscalibrated. By elimination — melt $7\times$ too high, formula validated, temperature realistic — the excess lives in the **cavity currents**: median 13 cm/s where ~ 2 cm/s would give the observed melt, i.e. $\sim 6\times$ too fast. Because the exchange velocity is $\gamma_T \propto |U|$, that fast circulation is exactly what inflates the melt; and as the shelves thin the currents speed up further — the escalation.

How to slow it without killing the climate response. The design constraint is to dissipate *momentum* with a flow-dependent operator, never to cap the diagnostic (velocity, γ_T , or melt) — a quadratic drag still lets the current find a slower equilibrium that shifts with forcing, whereas a cap freezes the response. The ice base is *not* free-slip: `cavity_momentum_fluxes` already applies quadratic drag $-\rho C_d |U|U$ at `nzmin`, correctly ordered (cavity elements skip the atmospheric-stress overwrite, then receive the drag). But it borrows the *seabed* coefficient $C_d = 2.5 \times 10^{-3}$ with the developer’s own flag in the source (“*need to check the sensitivity to the drag coefficient*”). The plan: split it into a separate, tunable `C_d_cavity` (default $= C_d$) and raise it — physically justified, since ice-shelf bases are hydraulically rougher than a flat seabed (basal crevasses, terraces, keels), and dedicated cavity models treat ice-base drag as a distinct tunable $\gtrsim$ seabed. Being quadratic it preserves the velocity’s climate sensitivity. Complementary flow-dependent levers if drag alone is insufficient: a vertical-viscosity floor in the cavity boundary layer, or Smagorinsky horizontal viscosity in cavities. *Not* to be done: raising the global C_d (over-brakes the open-ocean bottom boundary layer everywhere). Validation target: melt toward ~ 0.8 m/yr *while preserving* the spatial melt-forcing pattern ($r = 0.81$) and the warm-Amundsen/cold-FRIS contrast — brake the amplitude, not the geography. Caveat: part of the $2.5 \rightarrow 13$ cm/s jump over 1900 $\rightarrow$ 1901 is plausibly spin-up plus the increment shock, so retune against a settled multi-year state, not a transient.

On bias correction (a design question raised alongside): observation-anomaly (“delta”) ocean forcing is standard for stand-alone ice-sheet runs, and even cavity-resolving projections bias-correct their *atmospheric* forcing (Naughten et al. 2018 force FESOM with CMIP anomalies on an ERA-Interim climatology, control on reanalysis) — but it would not cure this run: the thermal forcing

here is already near-realistic, so correcting T/S addresses a problem we mostly do not have. The excess is in the *circulation*, upstream of both the melt formula and the water it is fed.

Superseded (2026-07-19): the “too-fast currents” diagnosis holds, but the proposed cure — a tunable ice-base drag C_{d_cavity} — does not: the standalone dissection below shows the fast currents are *injected by the increment*, not an under-damped equilibrium (chunk-1 and both year-long cold-start equilibria run ~ 2.6 cm/s at the stock drag). See §1.13; the drag plan is moved to Falsified.

1.13 The spin-up is the remapped restart, not the water — a standalone dissection (2026-07-19)

To test the “fast currents” hypotheses cheaply, a *standalone* FESOM harness (`build_sa`, `submesh_1901`, 20-day runs, ~ 2 min each) with a cavity median-column-speed metric. It reproduces the disease and then dissects it (Fig. 11).

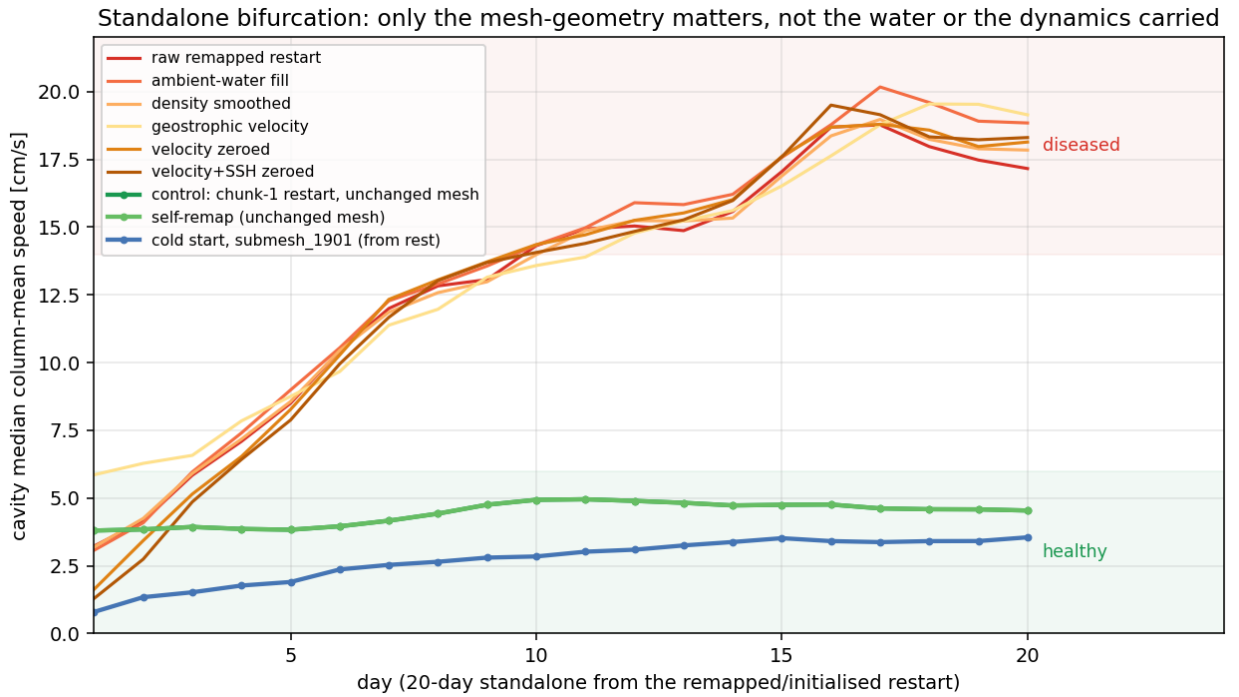


Figure 11: Standalone cavity median speed vs. day. The runs bifurcate cleanly by *what mesh geometry* they use, not by the water or the dynamics carried. **Healthy ($\sim 2\text{--}5$ cm/s)**: chunk-1 restart on the unchanged mesh (control), the same restart remapped onto its *own* mesh (self-remap, exonerating the remap machinery), and a cold start from rest on the changed submesh_1901. **Diseased (climb to $\sim 17\text{--}19$ cm/s)**: the raw remapped restart — *and every attempted fix of the carried state on that mesh*: cavity density smoothed, cavity refilled with ambient open-ocean water, geostrophic velocity imposed, velocity zeroed, velocity+SSH+hbar zeroed. All fast, all indistinguishable. The excess is density-independent and not fixable in any single carried field.

The elimination. (i) The harness is faithful — unchanged geometry stays slow, through the remap or not. (ii) The cavity *water mass is irrelevant*: chunk-1 water, harmonically smoothed water and a fresh ambient reservoir all spin up identically — baroclinic/density is not the driver. (iii) The *geometry is fine*: a cold start on submesh_1901 from rest settles to ~ 2.6 cm/s (1-year run, peaked $\sim$ month 3 then relaxed) — a healthy equilibrium lives on the new mesh. (iv) It is not the individually carried velocity, SSH or hbar (zeroing each $\rightarrow$ still fast) and `hnode` is already nominal. **Conclusion**: the injector is the *remapped restart being internally inconsistent with the new mesh*;

the geometry *change* is only the trigger that exposes it, not the cause.

Footprint. Matching submesh_1901 to the initial submesh by coordinate: of 2627 cavity columns, **1602 changed vertical structure but the remap flags only 410**. The classifier flags only columns that *gain* wet levels (need fill: ice base shallower or seabed deeper); the **1183 columns whose ice base got deeper** (thicker ice above → moved ice-loading/pressure boundary) are carried untouched. A density-independent, geometry-linked lead — not yet a proven cause.

Learning the target (in flight). Two 1-year cold starts with *identical* climatology and forcing, differing only in the mesh (1900 vs 1901), give two directly comparable equilibria. Their difference Δ_{equil} at the changed nodes is the *pure* ocean equilibrium response to the ice-draft change — the first-principles target a one-shot remap rule must reproduce (the old restart $\approx$ the 1900-mesh equilibrium; the modification should push it to the 1901-mesh equilibrium, derived, not copied).

1.14 The seam field is hnode (2026-07-20)

The equilibria then localised the seam to a single field. Building a rest state (velocity, SSH, AB, hbar all zero) with the *equilibrium* density and stepping it still spun up (median 19 cm/s) — so density is finally, unconfoundedly dead. That rest state differs from the healthy cold start in exactly one carried field: **hnode**, the ALE layer thickness. Swapping the remap’s **hnode** for the equilibrium’s (the *only* file that changes, by ≤ 0.73 m) flips the cavity from 18.6 to 4.5 cm/s, reproduced to two digits (Fig. 12). The changed-column differences sit a couple of levels below the moved ice base — which is why every density/velocity/SSH test walked past it. (*Initially* read as “differences entirely at the changed columns”; the correction below shows the swap also moved the open-ocean **hnode**, which carries most of the effect.)

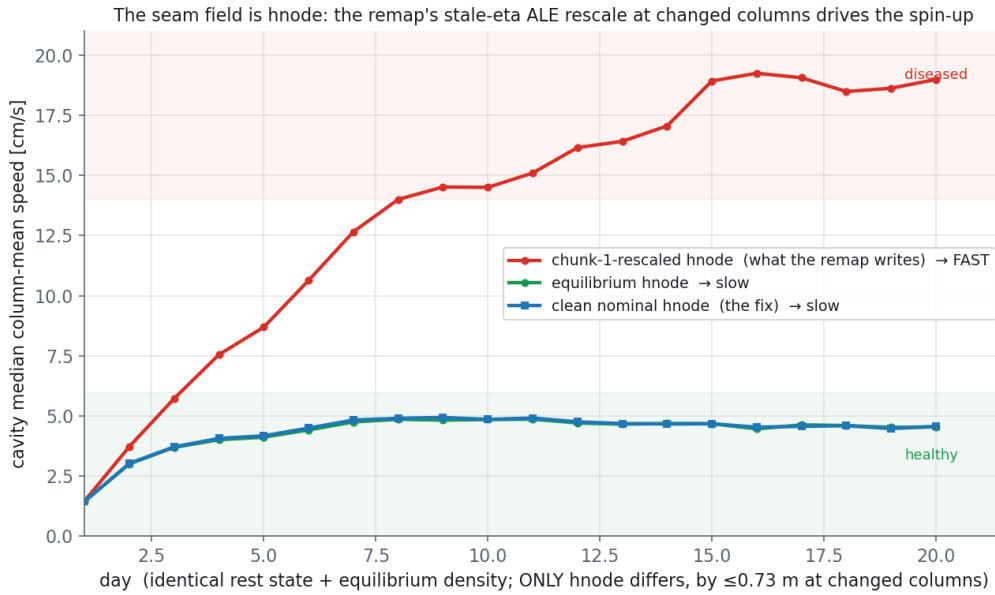


Figure 12: One field, one flip. Identical rest state and equilibrium density; *only* **hnode** differs, by ≤ 0.73 m at the changed columns. The remap’s chunk-1-rescaled **hnode** (red) spins the cavity to ~ 19 cm/s; the equilibrium **hnode** (green) and a *clean nominal hnode applied everywhere* (blue) both hold the healthy ~ 4.5 cm/s. *Caveat (added below)*: this swap also changes the *open-ocean hnode*, not only the changed columns — so it overstates what a changed-column-only fix achieves (that reaches only ~ 10 cm/s).

Mechanism. `remap_hnode` rescales every changed column so $\sum h = (z_{\text{bar}}(ul) - z_{\text{bar}}(nl)) + \eta_{\text{new}}$, using the *remapped chunk-1* η . At changed columns that η is stale — starkest at cavity→open (“closed”) nodes, where it is the under-ice depressed value (≈ -1.6 m) applied to a now-open column

— so the rescale *compresses* `hnode` (e.g. $10.0 \rightarrow 9.19$ m). That inconsistent thickness is exactly the “standing dh/dt source at $t=0$ ” the rescale was added to prevent; it seeds the barotropic spin-up. The rescale does the opposite of its intent because it trusts the carried η .

Fix, and the correction (2026-07-20 late). The `remap_hnode` edit — nominal z_{bar} thicknesses at the geometry-changed columns (including the ice-base-deepened ones the classifier calls “unchanged”) instead of the stale- η rescale — was implemented and regenerated end-to-end. In the standalone harness it *halves* the over-speed, $17 \rightarrow 10.3$ cm/s: a real, physically-clean improvement, but *not* a full restoration. Reaching ~ 4 cm/s requires nominal `hnode` over the *whole* ocean: the fully-fixed and the target restarts turn out *identical in the cavity* and differ only in the *open-ocean* `hnode`, and a cavity+ice-front-halo nominalisation does not help (11 cm/s) — only nominalising everywhere does. So the earlier “one field, one flip” was *confounded*: the clean swap carried the open-ocean `hnode` too, not just the changed columns. The residual is not the cavity fill but the **vertical-coordinate seam** at the ice front — `zstar` stretches the open-ocean layers by the free surface ($h_k = h_k^{\text{nom}}(D + \eta)/D$) while the rigid-lid cavity keeps them nominal; the remapped, `zstar`-stretched open-ocean `hnode` is what pushes the cavity from 4 to 10. This reframes “nominal everywhere” not as a global-reset artifact but as precisely what `linfs` (fixed layers, η a top-boundary diagnostic) does by construction — removing the seam without any `hnode` surgery. **Confirmed:** the raw `remap` (chunk-1 `hnode`, no surgery — the exact restart that spins to 17 cm/s under `zstar`) run with `which_ale='linfs'` does *not* spin up; it decays to ~ 1.7 cm/s median / 2.5 mean and holds. So the seam, not the cavity fill, was the mechanism, and `linfs` is the clean complete fix. Status: changed-column `remap_hnode` fix implemented (uncommitted, gives 10.3 — now superseded by `linfs`); the open question is whether `linfs` is acceptable model-wide for the coupled AWI-ESM3 open-ocean climate (a linearised free surface with mass-conservation trade-offs vs. `zstar`).

1.15 `linfs` in the coupled chain, and the open-cavity fill (2026-07-21)

The standalone finding was then taken into the real iterative coupling: a fresh run (`movcavlinfs`) with `which_ale='linfs'` from chunk 1, everything else identical, 1200s timestep. It validates the seam story end-to-end (Fig. 13). Four mesh increments (1900–1903) hand over with *no* cavity spin-up and *no* CFL blowup; the cavity node count is *preserved* (~ 2668 , vs the `zstar` run’s collapse $2647 \rightarrow 111$), the per-increment mesh change *shrinks* as the ice-ocean system settles ($402 \rightarrow 187$ nodes added), and cavity melt is stable and realistic — ~ 0.22 m w.e. yr^{-1} cavity-mean, ~ 1344 Gt yr^{-1} total (Rignot ~ 1325), with none of the $0.9 \rightarrow 5.5$ m yr^{-1} escalation seen under `zstar`. The vertical-coordinate seam was the mechanism; `linfs` removes it in the coupled system too.

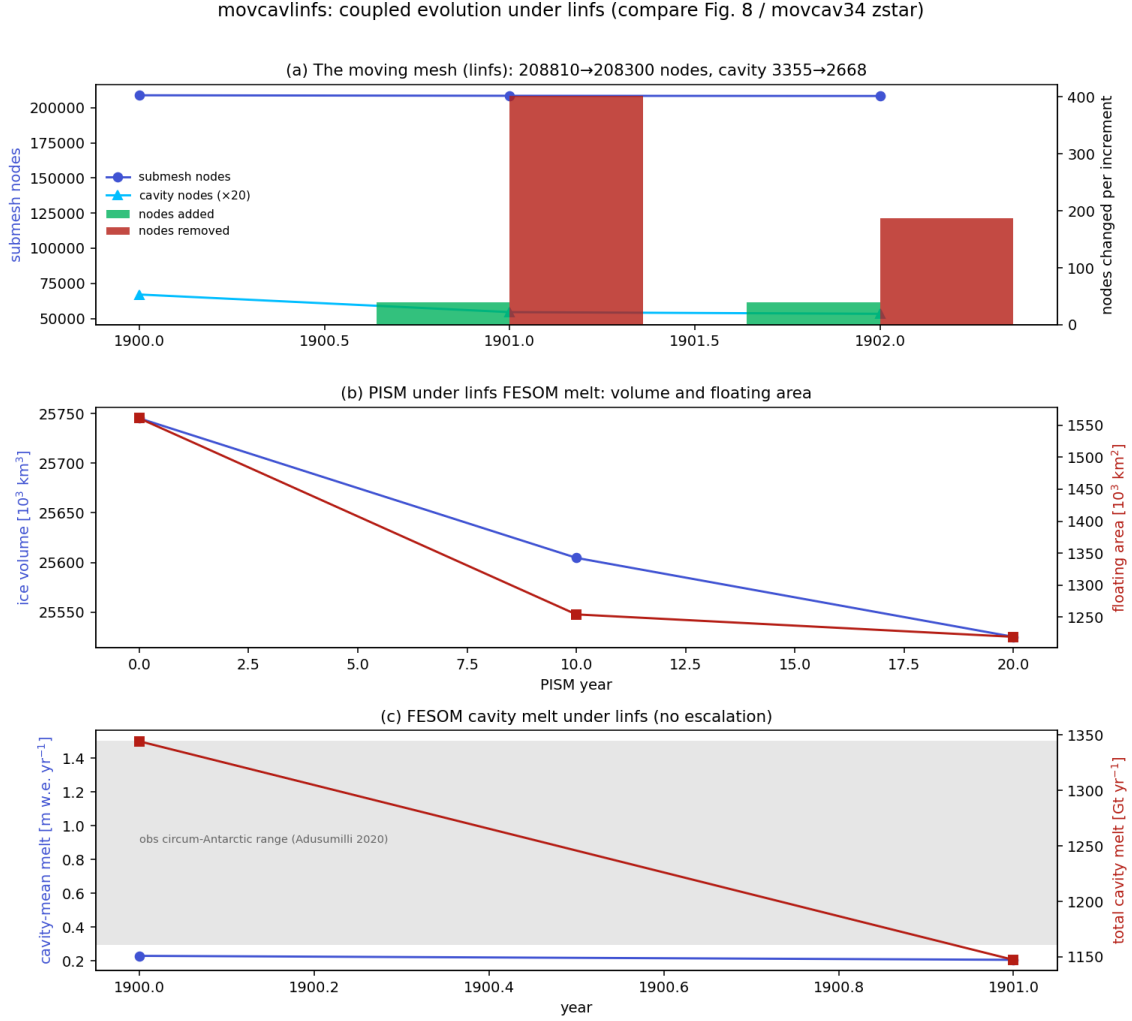


Figure 13: movcavlinfs coupled evolution under linfs (compare Fig. 8 / movcav34 zstar). (a) The moving mesh: node and cavity-node counts per increment, with nodes added/removed; the increment shrinks and the cavity is preserved. (b) PISM ice volume and floating area. (c) FESOM cavity-mean and total melt — stable in the observed circum-Antarctic range, no escalation.

The 1904 stop was a fill artifact, not the seam. Chunk 5 (1904) tripped FESOM’s fixed temperature sanity reject (“temperature becomes NaN or <-5.0 ”) at step 1 on a cavity node, $T = -5.35$. Not CFL, not spin-up. Two nested causes. (i) The remap carries each newly-iced node’s *own open-ocean* column into the new cavity, so where the ice front advances over warm water the ice base inherits it — up to $+6.6^\circ\text{C}$ of thermal driving at the new roof (Fig. 14a, red). (ii) A boundary-layer cap (`cap_new_cavity_temp`) that clipped those columns to the local freezing point *over-cooled* the deep part: at $\sim 4500\text{ m}$ the pressure-depressed $T_f = 0.0901 - 0.0575 S - 7.61 \times 10^{-4} z \approx -5.35^\circ\text{C}$, dragging real $\sim -0.5^\circ\text{C}$ deep water below FESOM’s -5 floor (Fig. 14a, blue).

Fix: seed newly-iced columns from the nearest existing cavity (user proposal). Instead of carrying a node’s open-ocean column (then capping it), each open→cavity column is seeded from the nearest column that is cavity in *both* meshes, depth-matched. This gives cold, cavity-appropriate ice-shelf water, regionally correct by nearest-neighbour (Amundsen-warm / FRIS-cold preserved), with the deep column left real — no cap needed. In the 1904 increment it seeds 204 recipients from 2326 donors and cuts ice-base warm columns ($> 0^\circ\text{C}$) from $61 \rightarrow 9$ and peak thermal driving from $+6.6 \rightarrow +0.8^\circ\text{C}$ (Fig. 14, green profile / right map); the residual nine are genuine warm East-Antarctic-coast cavities ($\leq +0.8$). A freezing-point floor (clamped to -4.9) is retained only as a last-resort backstop. No cap, no crash, deep water preserved.

Fix: seed newly-iced columns from the nearest existing cavity, not from open ocean

Ice-base warm columns ($>0^\circ\text{C}$): $61 \rightarrow 9$; max thermal driving $+6.6 \rightarrow +0.8^\circ\text{C}$. No cap, no crash (deep column stays real).

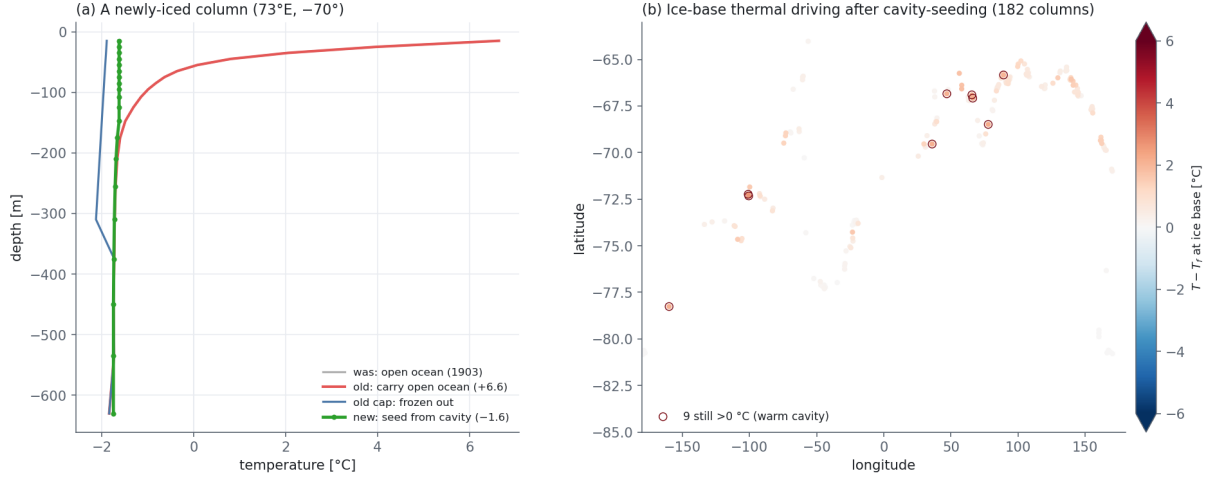


Figure 14: The open-cavity fill fix. (a) A newly-iced column: carrying its open-ocean water (red) puts $+6.6^\circ\text{C}$ at the ice base; the freezing cap (blue) over-cools the deep column; seeding from the nearest existing cavity (green) gives a real cold ice-shelf-water profile with the deep column intact. (b) Ice-base thermal driving $T - T_f$ over the 182 open $\rightarrow$ cavity columns after cavity-seeding — almost all near freezing; the nine circled ($>0^\circ\text{C}$) are physical warm-cavity donors.

1.16 Case closed: the full movcav37 cycle (2026-07-22)

With both fixes in place — `linfs` (coordinate seam) and cavity-donor seeding (open-cavity fill) — a fresh run `movcav37` was launched from chunk 1 and carried the complete decade 1900–1909: **ten coupled years, nine mesh increments, no spin-up, no blowup** (it passes 1904, where `movcavlinfs` had stopped). This closes the spin-up investigation.

The decisive metric is the cavity current itself (Fig. 15). The area-weighted cavity median column-mean speed holds **1.1–2.1 cm/s across all ten years** ($p_{90} \leq 5.2$), never leaving the healthy band — against the 13–17 cm/s the *same* remapped restart reached at every increment under `zstar`. The barotropic spin-up that this whole report chased is, by construction, gone: it was the vertical-coordinate seam, and `linfs` removes it.

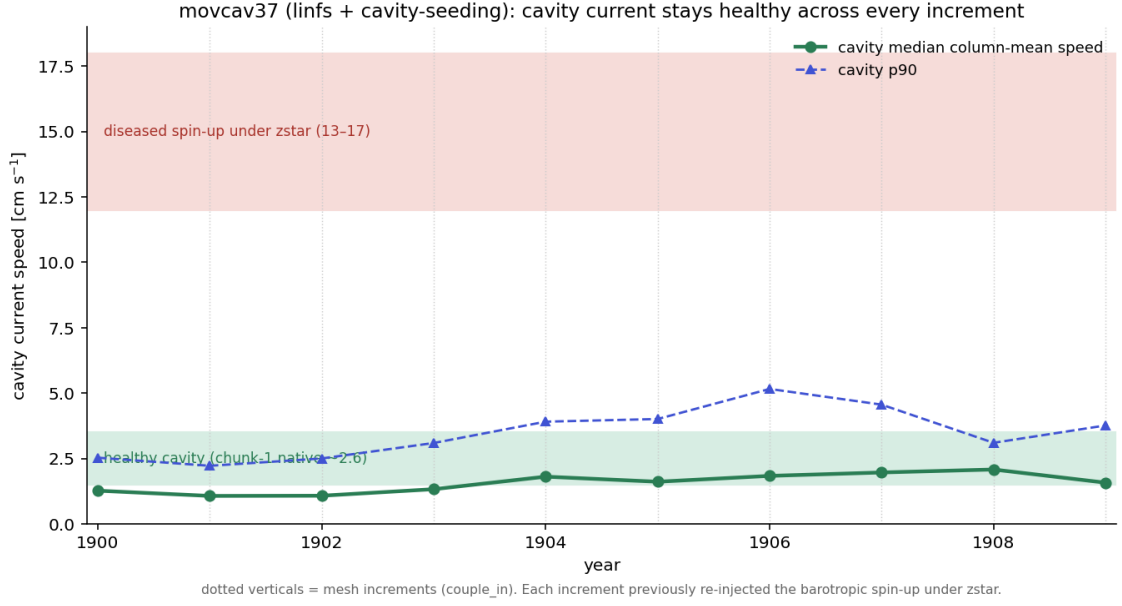


Figure 15: The spin-up, closed. `movcav37` cavity median (and p90) column-mean current speed per year; dotted verticals are the nine mesh increments. The current stays in the healthy band (chunk-1 native ~ 2.6 cm/s) at *every* increment — the 13–17 cm/s diseased spin-up under `zstar` (red band) never appears.

The coupled system is also *settling*, not diverging (Fig. 16). The cavity is preserved (3355 $\rightarrow$ 2318 nodes, a gentle thinning, not the `zstar` collapse to 111), the per-increment mesh change shrinks (405 $\rightarrow$ ~ 60 nodes), PISM volume and floating area decline smoothly with no shocks at the increments, and cavity melt stays inside the observed circum-Antarctic range (0.23 $\rightarrow$ 0.82 m w.e. yr $^{-1}$, 1329 $\rightarrow$ 1685 Gt yr $^{-1}$) — no trace of `zstar`’s escalation to 5.5 m yr $^{-1}$. The one feature to watch on a longer run is the mild *upward* melt trend; the flat 1–2 cm/s currents identify it as thermodynamic/geometric (shelves thinning, cavity warming slightly), not a dynamic spin-up, and it remains within observations.

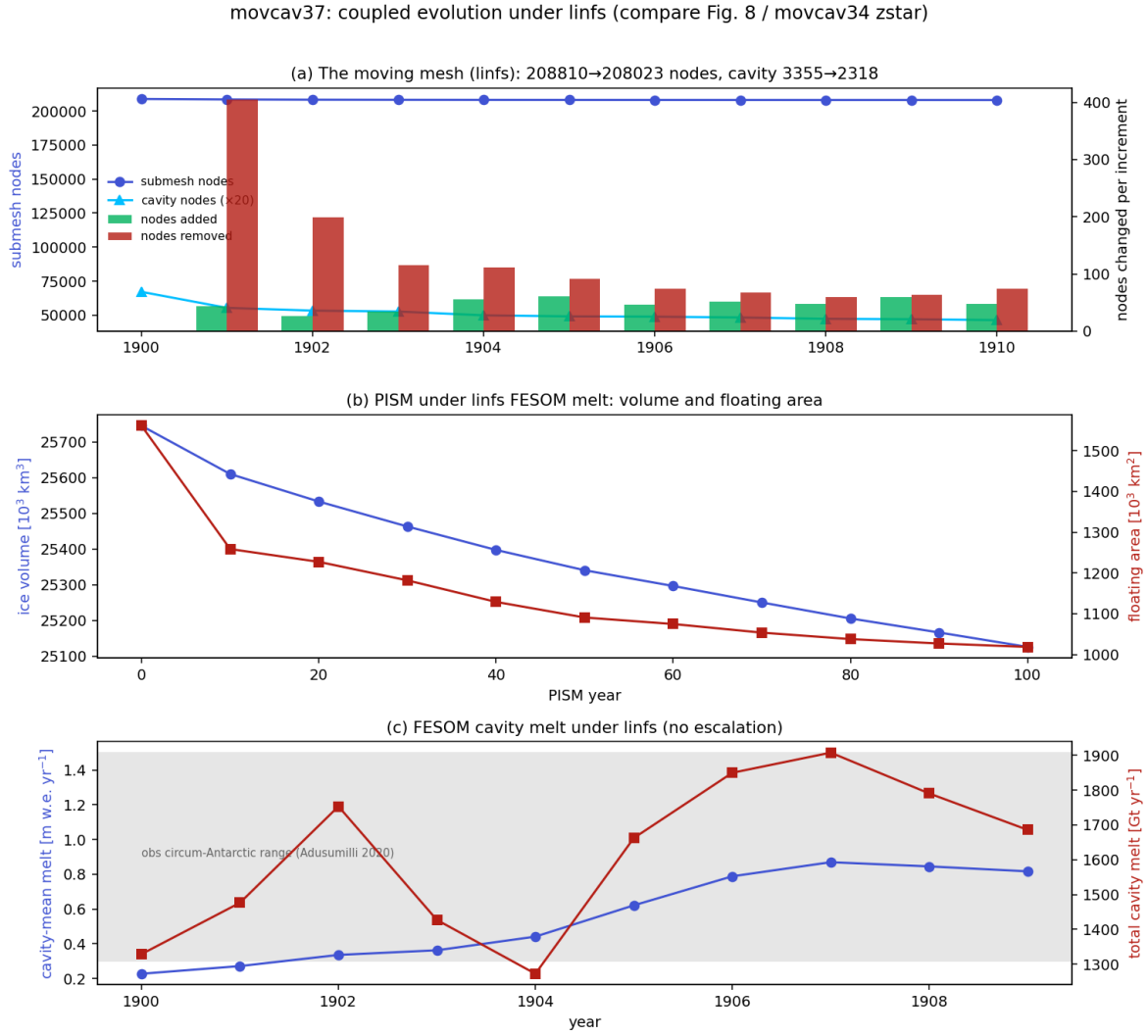


Figure 16: `movcav37` coupled evolution (compare Fig. 13 / `movcavlinfs`, now the full decade). (a) Moving mesh: cavity preserved, increments shrink. (b) PISM volume and floating area, smooth. (c) FESOM cavity melt, in the observed range, no escalation.

1.17 Sea ice and snow in `movcav37`: melt-back restored; the Ross lagoon (2026-07-22)

The open item of §2.3 — the SH summer melt-back and the coastal snow pile-up — is answered by `movcav37`. Diagnosed properly (sea ice lives on *open-ocean* surface nodes; cavity nodes carry spurious values and must be excluded), the thickness fields sit in the observed range in both hemispheres (Figs. 17, 18).

- **SH summer melt-back is restored.** The winter (Sep) pack is a coherent 0.4–1.0 m ring (thicker in the Weddell/Ross gyres); by summer (Feb) it retreats almost completely to thin Weddell/coastal remnants — the strong seasonal melt-back the reference had and the earlier `movcav` runs lacked. Hemispheric mean thickness ~ 0.6 m (obs ASPeCt/ICESat 0.5–1.0 m).
- **The coastal snow pile-up is gone.** SH snow is < 0.15 m over the pack with a coastal rim ~ 0.25 – 0.35 m, within the observed 0.1–0.4 m — not the 2–3.5 m band of the pre-fix runs. (The `linfs` + clean chunk-1 ice init + coupled slab combination did restore the melt-back; this closes the §2.3 question.)

- **NH is realistic:** 2–4 m ice against the CAA/Greenland thinning to the marginal seas (obs $\sim 1.5\text{--}3\text{ m}$), snow 0.15–0.35 m in winter (Warren $\sim 0.2\text{--}0.4\text{ m}$).

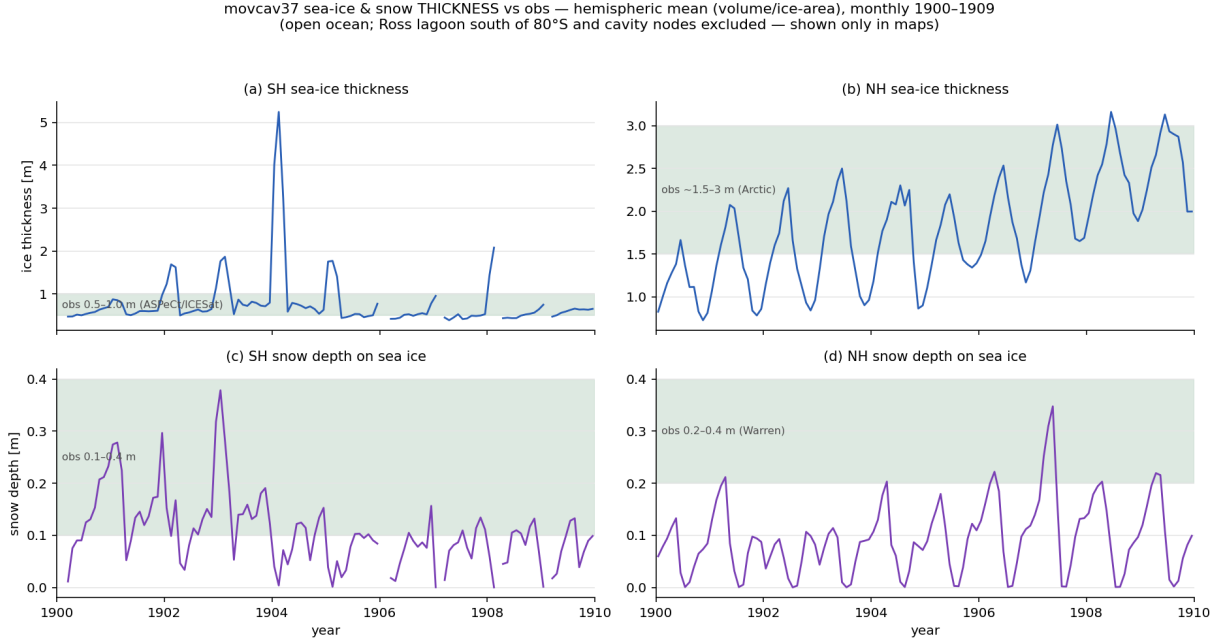


Figure 17: movcav37 sea-ice and snow thickness vs observations, monthly 1900–1909, hemispheric mean (total volume / total ice-area over ice-covered open ocean, $a_{\text{ice}} > 0.15$). Green bands = observed ranges. Cavity nodes and the Ross lagoon (open ocean south of 80°S) are excluded here and shown only in the maps. SH ice summer spikes are survivor bias (only thick coastal ice remains at the Feb minimum).

movcav37 seasonal sea-ice & snow thickness (5-yr clim 1905–1909) — SH top, NH bottom; ice (left) | snow (right)
 standard polar atlas view (SH: 0° at top, Ross Sea bottom; NH: 0° at bottom, Greenland bottom); black = present-day coastline. Ross lagoon = the saturated dark-red (~10 m) patch at the back of the Ross, SH panels.

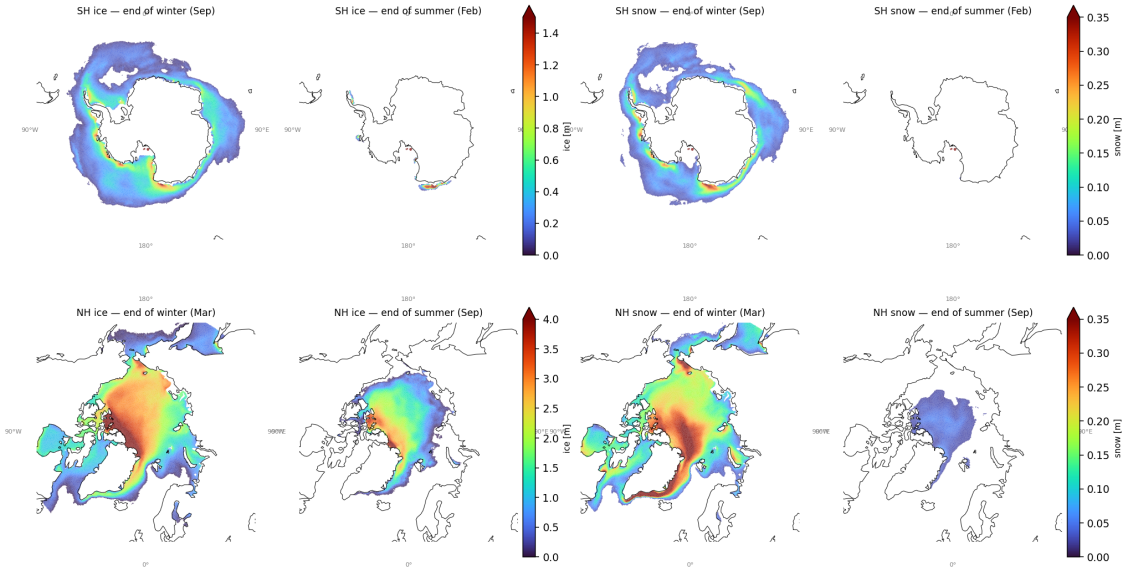


Figure 18: movcav37 seasonal (end-of-winter / end-of-summer) sea-ice and snow thickness, 5-year climatology 1905–1909, on the native grid (standard polar view, 0° meridian at bottom; grey = present-day coastline). SH top, NH bottom; ice (left pair), snow (right pair). SH ice melts back to near-nothing by February; SH snow is a thin coastal rim. The two dots off the Ross coast are the *Ross lagoon* artifact (below).

New open item — the Ross lagoon. The one unphysical thickness signal left is a small (~18-node) patch at the back of the Ross Ice Shelf (lon ≈ 180°, lat ≈ −83°) where PISM, over the run, thinned the shelf and retreated the grounding line until the classifier flipped those nodes from *cavity/grounded* to *open ocean* (verified node-by-node: `cavity(u1=12-16)` or land in 1900 → `u1=1` by 1909). FESOM then traps thick (~10 m), convergent, non-melting sea ice in the near-enclosed deep embayment — the dots in Fig. 18. It is a PISM-geometry artifact (likely over-thinning the back of the Ross, fed by excess FESOM cavity melt there), not a sea-ice-model problem; it is excluded from the aggregates above. Fix candidates: guard the classifier against ungrounding interior nodes far behind the shelf front, or address the excess back-of-Ross melt.

1.18 The coupled-slab switch was off for `develop/develop-cc`; the sea-ice climate validated on a clean decade (2026-07-23→27)

Rolling the coupled-slab work out from `develop-is` to the sibling setups `develop` and `develop-cc` exposed a silent, months-old configuration bug. The coupled slab is *hardwired in the FESOM build* (§1.8: `ice_thermo_cpl.F90` sets $Q_{\text{atmice}} = -a2ihf$, so the atmosphere owns the ice-surface temperature and its slab conduction must be fed the coupled ice thickness). OIFS is told to match through NAMCC: `LNEMOLIMTHK=.true.` (use the coupled thickness in the surface solve) and `LNEMOLIMTEMP=.false.` (do not overwrite the ice skin from the ocean model each hour). The OIFS defaults (`sumcc.F90`) leave *both* `.false.`. Only `develop-is` set NAMCC — and only because its *runscript* did, by hand. So `develop` and `develop-cc` silently ran with `LNEMOLIMTHK=.false.`: the `sit_feom→A_Ice_thickness` field added by the whole coupled-slab rollout was delivered over OASIS and then *discarded*, and the atmosphere conducted through a default slab thickness.

The signature was unmistakable once a clean 10-year `develop` run (`si10y`, TCO95–CORE3, 1990 GHG, fixed full mesh, no cavity) was diagnosed: with the switch off, the Arctic went *completely ice-free every June–October* and the winter pack carried half its proper volume (NH March volume

5.3; §2). Setting the three `NAMMCC` switches turned that into a physical sea-ice climate at the first attempt (Fig. 19): NH March volume $5.3 \rightarrow 27.4 \times 10^3 \text{ km}^3$ (obs $\sim 28\text{--}30$), NH September extent $0.0 \rightarrow 5.6 \times 10^6 \text{ km}^2$, SH February extent $0.0 \rightarrow 1.7$. The fix now lives in the setup config, not the runscripts (`awiesm3.yaml`, commit `0aca2487`), applied to all three `develop*` versions; tags keep the stock defaults because their component versions predate `sit_feom`.

This does not taint the `movcav` results. The `develop-is` runscripts set `NAMMCC` by hand throughout, so every `movcav` run in this report (§1.8 onward) already had `LNEMOLIMTHK=.true..` The bug lived only in the fixed-mesh siblings.

si10y ctrl vs tuned vs a270270 CMIP7 -- mean sea-ice thickness, native CORE3 grid, AFTER NAMMCC fix
ctrl/tuned = 1905-1909, ref = its yr 1355-1359. Arctic now survives summer and matches the reference spinup.

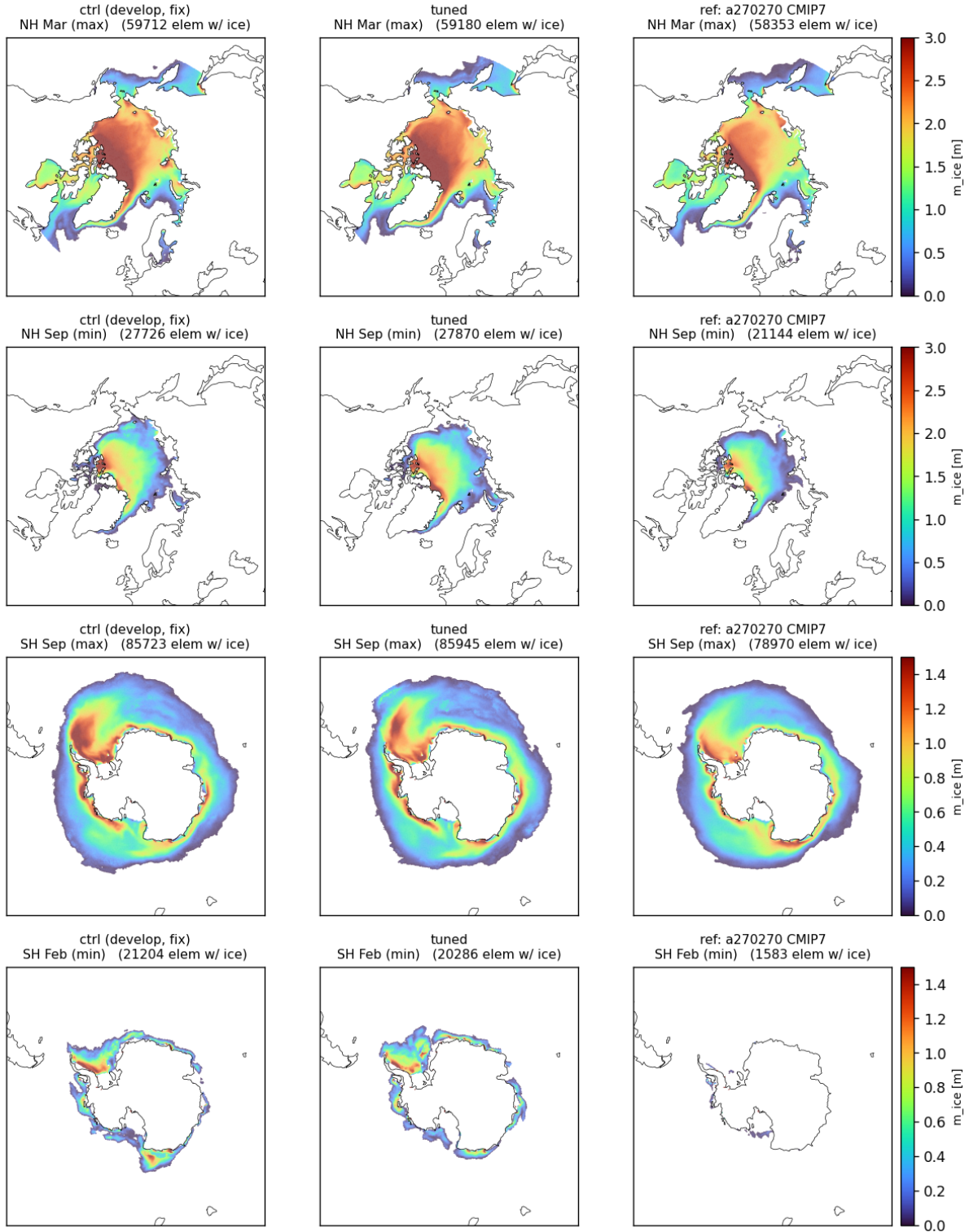


Figure 19: `develop` sea-ice thickness after the NAMMCC coupled-slab fix, native CORE3 grid, 5-yr mean of spin-up years 6–10, zoomed to the polar caps. Columns: control (fixed `develop`), the tuning experiment (§2), and an independent 500-year CMIP7 pre-industrial spin-up (a270270) at the same spin-up age. Rows: NH March (max), NH September (min), SH September (max), SH February (min); element counts with $a_{ice} > 0.15$ annotated. The NH September row — blank (“NO ICE”) before the fix — now carries a coherent central Arctic pack *exceeding* the reference (27.7k vs 21.1k elements). The large-scale fields match the reference spin-up hemisphere-by-hemisphere; the coupled-slab config is correct.

Quantitatively, against 1990s observations and the a270270 pre-industrial spin-up sampled at the *same* spin-up age (years 6–10 of each), last-5-year means:

metric	develop+fix (1990)	a270270 (PI)	obs
NH March extent [10^6 km^2]	16.1	15.4	~ 15.5 (NSIDC 1990s)
NH September extent [10^6 km^2]	5.6	4.7	~ 7.0
NH March volume [10^3 km^3]	27.1	23.1	$\sim 28\text{--}30$ (PIOMAS)
NH March mean thickness [m]	1.8	1.6	$\sim 2.3\text{--}2.6$
SH September extent [10^6 km^2]	21.4	20.3	~ 18.5 (NSIDC)
SH February extent [10^6 km^2]	1.7	0.0	~ 3.0
SH September mean thickness [m]	0.7	0.6	$\sim 0.9\text{--}1.2$

The fixed run is closer to observations than the reference on every row (Fig. 20). Two residual biases remain, both ordinary sea-ice tuning territory and neither addressed by the `whichEVP/melt-pond` tuning that was tried (it moved nothing): **NH ice is $\sim 20\text{--}25\%$ too thin** (thin pack, under-done September minimum), and **the SH seasonal cycle is too extreme** (over-grows in winter, over-melts in summer). A single-point thickness runaway ($\sim 52 \text{ m}$ at year 10) is present *identically* in the reference (which reaches 855 m by its year 500): a shared FESOM point-convergence artifact, not introduced by the coupling.

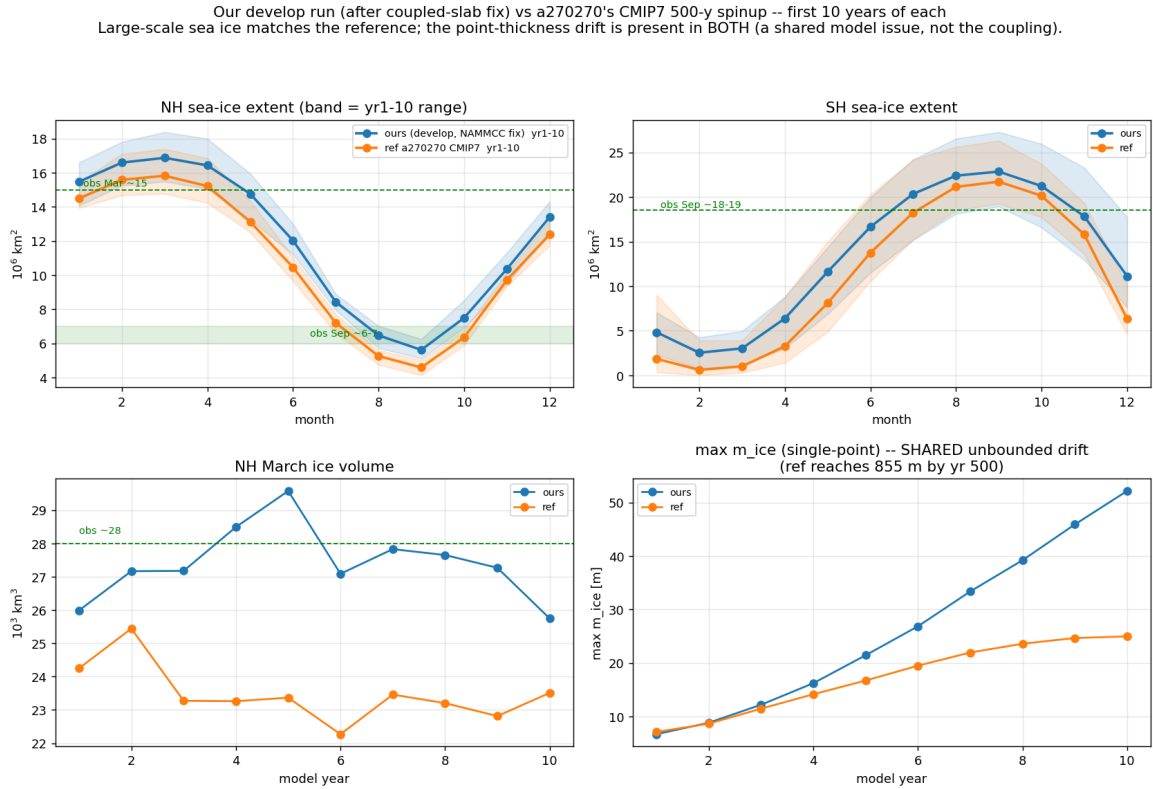


Figure 20: `develop+fix` (blue) vs a270270's CMIP7 pre-industrial spin-up (orange), first 10 years of each. (a,b) NH/SH seasonal extent, band = year-range; green markers = observed. (c) NH March volume; (d) single-point max thickness — the unbounded drift is present in *both* runs (shared model artifact). Large-scale sea ice matches the reference; where the two differ, the fixed run sits closer to observations.

A throughput trap surfaced while setting these runs up, worth recording because no short test catches it: with `oasis3mct.lresume=false` the coupling restart is rebuilt every exchange and the step rate decays $\sim 100\times$ over a run ($0.28 \rightarrow 11.4 \text{ s/step}$); `lresume=true` (from a matching restart set carrying `sit_feom`) holds it flat at $\sim 0.10 \text{ s/step}$. See Method notes.

1.19 The voskin:282 overflow, inventoried across the campaign and instrumented (2026-07-24)

The WAM fix (§1.5) closed one route into voskin_mod.F90:282, but the line kept killing later legs. Sweeping every error (72) floating overflow in the develop-is history gives eleven crashes, all on OIFS ranks and only ever at the rank extremes (OpenIFS decomposes north→south, so the extremes are the poles — never the mid-latitude ranks):

run(s)	chunk	hemisphere	crash site
movcav18/24/26/28/30/31	1 (1900)	both poles	cubasen:453, vsurf_mod:280
movcav19	5 (1904)	Arctic	cubasen:453
movcav33	2 (1901)	Antarctic	callpar:678
movcav34	13 (1912)	Antarctic	voskin_mod:282
movcav39	2 (1901)	Antarctic	voskin_mod:282
par03	2 (1901)	Antarctic	voskin_mod:282

Two families separate cleanly by chunk. **Chunk 1** crashes (cubasen/vsurf, both poles) are the pinned-skin sea-ice-thermodynamics instability, Bug IV — gone under the coupled slab. **Chunk ≥ 2** crashes (voskin/callpar, *Antarctic only*) are the mesh-increment family, and they persist.

voskin:282 is the *detector, not the cause*. The build is single precision (libsurf.SP.so, max $\sim 3.4 \times 10^{38}$); line 282 is the Langmuir number $Z_{LA} = [\rho_a/\rho_w \cdot Z_{UST}^2 / \max(u_{st}^2 + v_{st}^2, 10^{-8})]^{1/4}$, whose argument is $\sim 1.2 \times 10^5 \cdot Z_{UST}^2$ — it overflows only once the friction velocity Z_{UST} already exceeds $\sim 5 \times 10^{12}$. Upstream (lines 193–204) Z_{UST} is built from the surface stress PUSTR/PVSTR and a buoyancy amplifier Z_{WST2}/Z_{DU2} that divides by the skin temperature PTSKM1M; it is floored by ZEPUST but *never capped*. For par03 the crashing leg’s inputs were all checked and clean: the seven A096 fields of rstos.nc (including A_CurX/Y, which set the wind-relative-to-current stress) all physical, the regenerated ICMGG skt/sst/ci physical, and the A_SST coastline sanitizer (§1.4 lineage, commit e5d4796c) ran on that leg yet the crash still occurred. So the bad value is *generated inside the first timestep*, which is why file inspection has not found it.

Two diagnostic probes were added to voskin_mod.F90 (dump the column inputs when $Z_{UST} > 10$ or non-finite, and when the buoyancy Z_{BUO} blows up or PTSKM1M < 100 K), OIFS rebuilt (develop-is; VOSKIN-DIAG strings verified in libsurf.SP.so). The instrumented reproduction — par03 chunk 2 crashes ~ 60 s in — is **ready but not yet run** (it mutates the par03 experiment). Its output will separate the two candidate routes: huge PUSTR/PVSTR (stress arrived broken from the turbulence scheme) vs. a collapsed PTSKM1M/large fluxes (the buoyancy amplifier). *This is the current open front for the mesh-increment crash family.*

1.20 Two more crash faces: a mid-year FESOM flux blow-up and a leg-start numerical explosion (2026-07-30)

The voskin:282 front (§1.19) is OIFS-side and fires on step 1. Two *new* crash faces surfaced this week, both on the FESOM side, and neither is voskin:282.

(i) **A mid-year FESOM flux blow-up.** A colleague’s 17-cycle concurrent run (repro_movcav45) died in FESOM’s own check_blowup, not OIFS: cycle 19 (1918), mstep 14458 (mid-year, $\sim$ day 201), at an *Arctic* node (lat $+83.5^\circ$, lon -53°). temp hit 227°C , driven by a garbage surface heat flux hflux = $-4.0 \times 10^6 \text{ W/m}^2$ ($\sim 10^4 \times$ any physical air-sea flux) at an ice-edge node where sea ice vanished in a single step (m_ice 1.27 $\rightarrow$ 0). Only the surface layer blew up (every layer below normal, $-1.4 \dots -0.4^\circ\text{C}$) and CFL stayed < 1 throughout, so this is a *surface-flux injection*, not an interior/velocity instability. The submesh the coupler built for that leg was *clean* — cavity front stable at ~ 704 km every year (1910–13), matching PISM’s ~ 690 km — so the PISM→FESOM geometry conversion (mesh_flags.py + fesom_submesh.x) is *correct*; the bad flux is delivered by the

atmosphere/OASIS, not made by a mesh error. Distinct from voskin:282 (OIFS overflow, Antarctic, step 1); this is FESOM, Arctic, mid-year. The run’s output was monthly / yearly-split with NLOGPRT=1 and EXPORTED (not EXPOUT), so nothing flushed before the crash and the delivered flux at that step could not be recovered from disk — which set up the instrumented rerun.

The instrumented crash-hunt build (fgdbg02). To catch the flux *at ingest*: a [FLUXGUARD] added to check_blowup (write_step_info.F90) that, every step, prints mstep, node, lon/lat, qso/qsi/qsr, a_ice/m_ice for any node whose ingested oce_flux_h/ice_flux_h exceeds 5000 W/m^2 or is NaN; a *daily, daily-split* diagnostic stream of the atmosphere→ocean/ice flux components (qsi/qso/qsr plus the full fh decomposition fh_sen/fh_lat/fh_radtot/fh_swr/fh_lwr/fh_lwrout and uwice/vwice) with split_freq=1d so each day lands on disk before a mid-year blow-up (the previous config’s monthly, yearly-split output is exactly why nothing survived); and FESOM omp_num_threads=1 for bit-reproducibility, so a single crashing leg can later be re-run deterministically. Run for 100 yr because the blow-up timing is random.

(ii) **A leg-start numerical explosion — a *third*, distinct face.** fgdbg02 blew up at cycle 7 (1906), but at mstep 1 — the very first timestep of the leg — at an *Antarctic ice-shelf* node (lat -72.9° , lon -100° , Amundsen Sea). temp went $2.8 \rightarrow -3.7 \times 10^{243}$ in one step (a full numerical explosion, top ~ 17 levels, not a physical 227°C overshoot); hflux= -90.5 W/m^2 (*normal*), hnode/Kv/W/ CFL_z all normal, and [FLUXGUARD] stayed **silent**. The field that stood out in the dump is the SSH-solver right-hand-side: ssh_rhs= 7.1×10^6 (from -208 the previous step), which is what this entry originally read as an *external-mode (SSH) equation* blow-up. ~~The couple-in-hands FESOM a transport-geometry mismatch that detonates the SSH solve on step 1.~~ That reading did not survive opening the blow-up dump: ssh_rhs $\sim 10^6$ is the *ordinary* value at a cavity-front node and the real fault is elsewhere entirely (§1.21). What this entry does establish and what still stands: the run is on which_ale=’linfs’ (step_per_day=72, a 1200s step), the scheme §1.15 showed *preserves* the cavity where zstar collapses it, so face (iii) is **not** the zstar seam collapse and linfs does not cure it; it is *not* advection, *not* hnode (§1.14, fixed; clean here), and *not* the ALE-mode issue of §1.15. The daily stream captured nothing (crash before day 1 closed) and the guard correctly stayed silent, validating the instrumentation while telling us this crash is not flux-driven. The omp=1 build makes it bit-reproducible on one leg, which is what made §1.21 possible.

Three faces now, to keep separate. (i) voskin:282: OIFS single-precision overflow, chunk ≥ 2 , Antarctic, step 1. (ii) mid-year FESOM flux blow-up: Arctic, ice-edge, a garbage OASIS-delivered heat flux. (iii) leg-start FESOM SSH-RHS explosion: Antarctic ice-shelf, *normal* flux, mstep 1, ssh_rhs= 7.1×10^6 under linfs. Faces (i) and (iii) share step-1 timing and the Antarctic cavity region and may be the same underlying couple-in seeding; (ii) is a separate atmosphere-side flux corruption. The omp=1 build makes face (iii) bit-reproducible on one leg — the cheapest next target.

1.21 Bug VIII — an unassigned array element in the GM streamfunction (2026-07-31→08-01)

Face (iii) is solved, and it is not an SSH problem. The 2 GB fesom.1906.oce.blowup.nc that check_blowup writes — a full 3D dump of ssh_rhs, d_eta, u, v, u_rhs, helem, hnode, w, cfl_z, temp, salt at the crash step — had never been opened. It settles the face in one pass.

The SSH framing dies first. ssh_rhs= 7.1×10^6 is an unremarkable value here: 1461 nodes exceed 10^6 , 506 exceed 10^7 , the global max is 5.97×10^7 ($8\times$ the crash node), and the crash node ranks **143rd of the 184** pinched open nodes. d_eta stays within $[-1.76, +0.07] \text{ m}$ everywhere — the CG solve absorbs the large rhs. There is no SSH-solver explosion. The large ssh_rhs at the front is itself structural: eta is identically 0 at all 2416 cavity nodes (FESOM’s own convention, present in the native end-of-year restarts too) while the open ocean south of 60°S sits at -1.58 m , so each of the 1311 mixed cavity/open elements carries a $\sim 1.59 \text{ m}$ eta step and a $-g\nabla\eta dt \approx 1.5 \text{ m/s}$ velocity increment. Median $\max|u_rhs|$ over those 1311 front elements is **1.01 m/s** against 0.006–0.007 for open-ocean and full-cavity elements; the crash node’s elements (1.11–1.25) sit between the p50 and

p90 of that population. They are typical front elements.

The needle. Exactly **three** elements in 404607 carry an interior exact 0.0 in `u` where `v` is non-zero (none in `v`, none in `helem`), and they touch exactly the five nodes whose tracers exploded. The same test over the campaign’s thirteen saved restart states across seven cycles returns **zero**. A probe called at seven points of the timestep fired only *after* `solve_tracers_ale` — silent after `compute_vel_rhs`, `viscosity_filter`, `impl_vert_visc_ale`, `update_vel` on both sides of `exchange_elem`, `compute_hbar_ale` and `vert_vel_ale`.

Root cause. `fer_solve_Gamma` (`oce_fer_gm.F90`) solves the Gent–McWilliams streamfunction over `nzmin=ulevels_nod2D_max(n)` to `nzmax=nlevels_nod2D_min(n)` — max and min over the node’s *surrounding elements*. A node touching both a deep-cavity element and a very shallow one inverts that range. At node 19343 (elements with `ulevels` = [7, 1, 1, 1] and `nlevels` = [18, 18, 14, 5]) it is `nzmin`= 7, `nzmax`= 5. Both sweeps then run zero times, but

```
tr(:,nzmax) = tp(:,nzmax)
```

executes unconditionally and publishes an **unassigned element of the local automatic array** `tp` straight into `fer_gamma`. `fer_gamma2vel` divides gamma differences by `helem` to build the bolus velocity, giving the measured `fer_uv(1)` = $\mp 1.636 \times 10^{251}$ at levels 4 and 5 — equal magnitude, opposite sign, zonal component only, with `fer_uv(2)` $\sim 10^{-4}$ and `helem`= 10m both healthy, exactly the signature of one corrupted gamma at the level-5 interface. `solve_tracers_ale` then adds `fer_uv` to `UV`, **advects the tracers with it** (`temp`→ 10^{250} , `salt` pinned at its 3/45 bounds) and subtracts it back; $(x + y) - y$ with $|y| \gg |x|$ returns exactly 0.0, which is the half-zeroed `u`. The identical add/subtract pair on `Wvel` with `fer_Wvel` five lines below produces the `W`= 0 holes. So the chain runs the *opposite* way to the first reading: the exact zeros are residue of the event, not its seed.

Why now, after a decade of FESOM. The trigger needs a node touching both a deep-cavity element and a very shallow one — a cavity front on steep bathymetry, which the moving cavity manufactures and relocates every cycle. The predicate `max(ulevels) > min(nlevels)` over a node’s elements finds 1/2/1/2/0/2 such nodes in submeshes 1901–1906 and 2 in the base mesh, so the configuration recurs without always being fatal: `tp(:,nzmax)` is whatever the stack holds, usually benign, occasionally 10^{252} . That is the 1-in-hundreds behaviour of §1.20, and it is why the identical source crashed at `mstep` 1 under `-O3` but ran clean under `-check bounds -check pointers -check uninit -g`. `-check bounds` cannot catch it: `fer_gamma` is written in bounds; the defect is the *read of an unassigned value*. Requires `Fer_GM=.true.` (set here) and, for the degenerate range, `use_cavity=.true.`

Fix and verification. A node with no interior range carries no GM transport, so zero its column and skip:

```
if (nzmin >= nzmax) then; tr(:, :) = 0.0_WP; cycle; end if
```

The leg had died at `mstep` 1 on every previous attempt, bit-identically; with the fix it ran to day 27 / step 1920 with both detectors silent and no blow-up. Upstream as [FESOM/fesom2 PR #960](#) (branched off `main`, which carries the same defect), cherry-picked onto `awiesm3-implicit-ice-surftemp` as 26907ba9.

And then it just ran (2026-08-01). Restarted from cycle 7, `fgdbg02` chained itself unattended to **cycle 25 (1924) — 18 further coupled years, 0 blow-ups, 0 check_blowup STOPS**, couple-out → `PISM` → couple-in → compute with no intervention. That is the real confirmation the single day-27 run could not give: the degenerate-node configuration recurs across submeshes, so 18 fresh meshes and 18 fresh couple-ins is a far stronger test than one leg, and it held on all of them. Incidentally this also carried past cycle 19 (1918), where the colleague’s `repro_movcav45` hit face (ii) — encouraging but not conclusive, that face being random in timing.

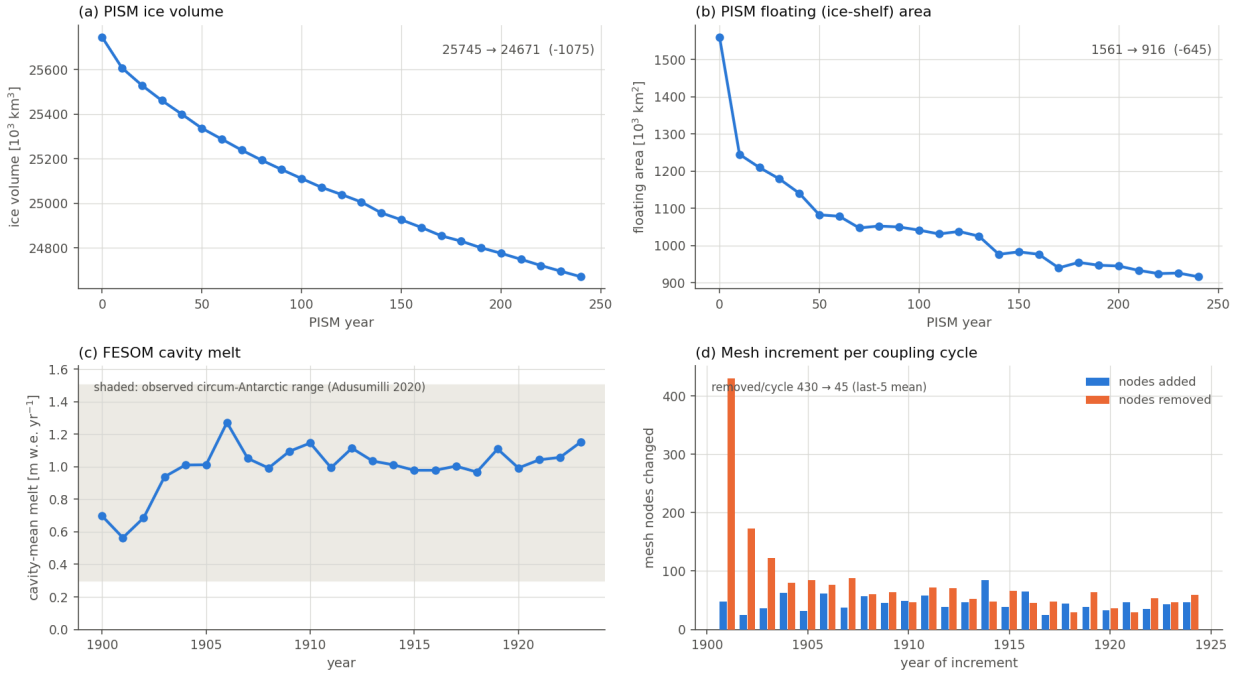


Figure 21: The coupled system equilibrating over 24 coupled years / 240 PISM years, post-fix. (a) PISM ice volume $25\,745 \rightarrow 24\,671 \times 10^3 \text{ km}^3$ and (b) floating area $1561 \rightarrow 916 \times 10^3 \text{ km}^2$: both still declining but visibly decelerating, the shelf area flattening after $\sim$ PISM year 150 — the system is *approaching* a steady state, not yet at one. (c) Cavity-mean melt runs $0.70 \rightarrow \sim 1.05 \text{ m w.e./yr}$, inside the observed circum-Antarctic band (shaded) — no escalation, which is what the **zstar** runs failed to do (§1.16). (d) The geometry increment is what equilibrates most clearly: nodes removed per cycle decays $430 \rightarrow 45$ (last-5 mean) while nodes added stays flat at 25–85, i.e. successive submeshes converge. Panels use one y-axis each by design; the earlier timeline figures’ twin axes invite misreading of exactly these trends.

TAINTED: cycles 1901–1905 ran with this active. Wherever a degenerate node existed and the stack word was small-but-non-zero, the GM bolus transport there was junk without crashing, so the GM contribution in those legs is unreliable even though they completed.

2 State of the cycle and of the climate (2026-08-02)

2.1 What is proven

- **Chunk-1 is quantitatively healthy and, under the coupled slab, crash-free:** January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; kilowatt flux events at real ice tiles: zero.
- **The cycle runs unattended, at length.** repro_movcav45 has carried **57 coupled years / 570 PISM years** (1900→1957) and fgdbg02 24, including **18 consecutive legs after the Bug VIII fix** with zero blow-ups. Everything the coupling regenerates per leg — submesh, partition, exchange weights, remapped restart, OIFS ICMGG/plit — has now been exercised 57 times without intervention. This supersedes the increment-test and live-mesh-change milestones of §1.10.

- **linfs cures the remapped-restart spin-up** (§1.14, §1.15), with 57 cycles behind it rather than one decade.
- **Cavity melt sits in the observed band and does not escalate:** $0.70 \rightarrow \sim 1.05$ m w.e./yr over fgdbg02’s 24 cycles, against the observed circum-Antarctic 0.3–1.5 (§1.21, panel c).
- **Bug VIII is fixed and upstream** (PR #960), with two further out-of-bounds reads found alongside it and one XIOS ice-mask fix (PR #961).

2.2 Taint ledger — what must not be quoted

- **All PISM-side results from 2026-07-17 to 2026-07-30, and fgdbg02 entirely.** The FESOM→PISM sub-shelf mass flux was delivered with *inverted sign*: FESOM’s `fw` (melt < 0) was passed through unflipped, and PISM’s `shelfbmassflux` is positive = melt, so PISM accreted where the ocean melted. Measured in fgdbg02: `basal_mass_flux_floating` = $+694 \text{ kg m}^{-2} \text{ yr}^{-1}$, 90.5 % positive (PISM’s own attribute: “positive = ice gain”), against FESOM’s own ~ 1.0 m w.e./yr of melt in the same cavities. Fixed by 33f71c607 (`factor` = -1 on the `fw` partable entry), which `repro_movcav45` has had throughout ($-450 \text{ kg m}^{-2} \text{ yr}^{-1}$, 14.5 % positive — correct). This kills the ice-volume and floating-area panels of §1.21 and the “gentle geometry response” of §1.10.
- **Cycles 1901–1905 of fgdbg02** additionally ran with Bug VIII live: wherever a degenerate node existed and the stack word was small-but-non-zero, the GM bolus transport there was junk without crashing.
- **The eleven hygiene-amputated legs** (§1.11), already noted.
- **Any earlier figure using the node-area helper.** Its plane-projection formula exploded on the ~ 1000 elements spanning $\pm 180^\circ$, inflating the global mesh area from 361 to $860 \times 10^6 \text{ km}^2$. Fixed to spherical excess across all plotting scripts (2026-08-02); the fgdbg02 cavity melt moved $0.21 \rightarrow 0.70$ at its start as a result.
- **Not tainted:** the FESOM-side melt diagnosis — cavity currents $\sim 6\times$ too fast, the `hnode` seam, the `linfs` cure (§1.12–1.14) — was measured inside FESOM and is unaffected by any of the above.

2.3 Open items

1. **fgdbg02 must be redone with the sign fix.** Its 240 PISM years are spent as science; they remain a valid crash-hunt record for Bug VIII.
2. **Southern-Hemisphere sea ice — reframed.** The old “we retain too much summer ice” item is *withdrawn*: it compared against the v3.4 CMIP7 reference, which carries roughly *half* the observed Antarctic sea ice. SH extent (10^6 km^2 , last 5 yr of each run, against ~ 18.5 max / ~ 3.0 min observed):

	max	min	
repro_movcav45 (57 yr)	22.97	2.56	too much winter ice
fgdbg02 (24 yr)	18.86	0.32	melts back too far
v3.4 CMIP7 reference	10.08	0.01	<i>not a usable yardstick</i>

The two runs err in *opposite* directions and neither matches observations. Why they differ is open; the sign fix is a candidate lever (correct-sign melt freshens and stratifies the shelf front, which preserves summer ice).

3. **voskin:282** — **dormant, not fixed**. Last fired in `repro_movcav45` at year 1919 (`voskin_mod.F90:318`); 38 cycles clean since, and never in `fgdbg02`. Instrumentation is in place (§1.19).
4. **The mid-year Arctic flux blow-up** (face (ii), §1.20) has not recurred and was never reproduced.
5. **The single-point thickness runaway**: present *identically* in the reference CMIP7 spin-up, so a shared FESOM point-convergence artifact rather than a coupling problem.

3 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured -45°C) when columns deepen	take water from the nearest old node wet at that depth	eef4f6db
Sub-ice η/hbar kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	b17fb5c8
First mesh-change leg read the mother mesh as cavity-free (<code>cavity_nlvls.out</code> missing from the <code>mother_as_old</code> links) $\Rightarrow$ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk <code>couple_in</code> rerun promoted its own submesh to <code>previous_submesh</code> (old = new $\Rightarrow$ zeros again)	rerun guard	b9c1e370
OASIS GSSPOS conservation blowup on near-zero target residuals	<code>gauswgt_gsmart</code> transform	3e94d4da
<code>namcouple feom</code> dimension never actually patched (env var from another subjob's shell)	fixed variable passing	3cf29e9e
<code>fesom2ice</code> loaded the static full mesh against submesh output	load the run mesh	63cfcc91
<code>previous_submesh</code> read but never created	created; plus restart size guard	3b44a704
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	<code>.coupling_failed</code> sentinel honoured by observe	7953585b
<code>rstos</code> size guard read the wrong dimension	fixed	61c8ed8a
<code>ocp-tool</code> flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51
Non-short-circuit <code>.and.</code> read <code>ulevels(0)</code> at every boundary edge (<code>ice_maEVP.F90</code> , both EVP variants); inside <code>if (use_cavity)</code> , so a garbage value decided whether sea-ice velocity was zeroed at the cavity-ocean edge	guard <code>edge_tri(2,ed)>0</code> separately	PR #960

Bug	Fix	Commit
Non-short-circuit <code>.and.</code> read <code>angles(0)</code> in the XIOS cell-corner insertion sort (<code>io_xios.F90</code>); harmless at -O3, aborts any bounds-checked build at init	split the loop test	PR #960
<code>alb</code> missing from <code>io_xios_is_ice_field</code> although declared <code>detect_missing_value="true"</code> $\Rightarrow$ sea-ice albedo time-means averaged over open water	add 'alb' to the ice-field list	PR #961

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

4 Falsified claims (kept, struck through)

- ~~Geostrophically-unbalanced remapped state; initialize changed cells in thermal-wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.
- ~~The sub-ice ssh-BC violation is the blowup.~~ Real bug, fixed, run still died.
- ~~The per-leg PISM increment is gentle (≈ 22 m mean).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).
- ~~The salt-plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.
- ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.
- ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.
- ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §2), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.
- ~~Fragile ≤ 2 -element coastal slivers cause the Ross-front blowup.~~ A control census killed it: the meshes that ran clean full years carry the same ~ 3450 -node fragile count as the crashed one. The cause is increment magnitude, not coastline fragility (§1.11).
- ~~The cavity melt gain is $\sim 10\times$ too steep (39 vs 1–4 m/yr/K).~~ The 1–4 figure is an integrated-shelf number, not the local exchange sensitivity; the standard three-equation scheme gives ~ 40 m/yr/K locally, so the measured 39 is correct. The melt formula is standard and reproduces the model's melt ($r = 0.91$); the excess is in the cavity *currents* (§1.12).
- ~~The fast cavity currents are an under-damped equilibrium, cured by a tunable ice-base drag ϵ_d cavity.~~ At the stock drag, chunk-1 and *both* year-long cold-start equilibria run at the healthy ~ 2.6 cm/s; the fast currents are injected by the mesh increment, not a drag deficit (§1.13).
- ~~The increment's geometry change injects the spin-up.~~ A cold start from rest on the changed mesh settles to ~ 2.6 cm/s — the new mesh is fine; the carried *remapped restart* is what is inconsistent with it (§1.13).

12. ~~The barotropic margin force F_x (eta+sea-ice) drives the spin-up.~~ The slow control carries the same F_x distribution ($p90 \sim 7 \times 10^{-4}$) as the fast runs; the step-1 force does not discriminate slow from fast (§1.13).
13. ~~The leg-start crash is an SSH-RHS explosion: the couple-in-hands FESOM a transport-geometry mismatch that detonates the external-mode solve.~~ `ssh_rhs` = 7.1×10^6 is ordinary at a cavity front — 506 nodes exceed 10^7 , the crash node ranks 143rd of 184 pinched nodes, and `d_eta` stays ≤ 1.76 m everywhere because the CG solve absorbs it. The fault is in the GM streamfunction (§1.21).
14. ~~The pinched cavity-front peninsula tip drives the blow-up: remapped velocity has nowhere to go through the pinched faces.~~ The pinch is the normal front geometry and its `ssh_rhs` signature is unremarkable at the crash node. The pinch *is* implicated, but through `ulevels_nod2D_max` inverting the GM solve range, not through trapped transport (§1.21).
15. ~~The exact zeros in `u` and `w` seed the tracer explosion via a singular vertical operator.~~ `vert_vel_ale` runs *before* `solve_tracers_ale` and the probe after it is silent, so `w` was computed from a clean `uv`. Both zero fields are residue of the $(x+y) - y$ cancellation in the GM bolus add/subtract pair — the chain runs the other way (§1.21).
16. ~~The fix belongs in the restart-remap (`mo_remap_fields.F90`).~~ The three affected elements come out of the remap clean and geometrically identical to the previous cycle (unchanged `ulevels/nlevels`, none new); `area/areasvol`, the partition and its halo comm arrays are all self-consistent. The defect is in FESOM's GM solver, exposed — not created — by the moving cavity (§1.21).

5 Method notes

1. **Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.
2. **Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I's diagnosis for a week).
3. **“Same code as the stable reference” localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.
4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.
5. **Beware silent fallbacks** (Bug V's `isfile()` miss; the `-ocean` given silent fallback to `th`). A fallback on a nonexistent path should say so. The NAMCC bug (§1.18) is the same shape: OIFS silently ran on default switches and discarded a coupled field for months, visible only in the sea-ice climate.
6. `oasis3mct.lresume=false` **silently destroys throughput.** With it false the coupling restart is rebuilt every exchange and the step rate decays $\sim 100\times$ over a run ($0.28 \rightarrow 11.4$ s/step) — no single-leg or one-month test is long enough to see it. Use `lresume=true` from a restart set that carries the current coupled fields (the shared pool `rst*` predate `sit_feom` and cannot be used with `lresume=true` until regenerated).
7. **In a coupled run FESOM is not forcing-driven.** `namelist.forcing`'s JRA55-do paths are vestigial (standalone only); the ocean surface fluxes come from OIFS through OASIS, so the climate is set by the OIFS GHG year (`NCMIPFIXYR`: 1990 for `develop` here, 1850 for the a270270 reference), not by the forcing files.

6 Assets

- Send/receive flux instrumentation: `SENDDBG` (OIFS `A_Q_ice` decomposition + tile state), `ICEDBG` (FESIM received flux + state at cold nodes), `CPLDBG` (OIFS ingest bounds) — all log-only, strip before merging.
- `oasis_expout_o2a` runsript flag: every o2a exchange dumped to netcdf ($\sim 20\times$ slowdown; crash-window probes only).
- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields index-mapped to the submesh; `map_nod.out` makes such remaps exact.
- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.
- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).
- Face-(iii) probe set (`notes/ssh_rhs_probes/`): `blowup_uvrhs.py`, `eta_seam.py`, `front_census.py`, `uv_holes.py`, `uniqueness.py`, `inverted_range.py` (the `max(ulevels) > min(nlevels)` mesh predicate), `read_areas.py`, `partition_check.py`, `comm_symmetry.py`. Working notes in `notes/ssh_rhs_leg_sta`.
- `fesom_mesdiag` (a standalone utility already in the FESOM tree, `BUILD_MESHDIA=ON`) dumps `area/areasvol` per level without running the model — how §1.21 cleared the control-volume hypothesis.
- The crash-step blow-up dump `fesom.1906.oce.blowup.CRASH1_KEEP.nc`: a full 3D state at `mstep 1`. `check_blowup` writes one on every blow-up and it is routinely ignored; opening it is what solved face (iii).